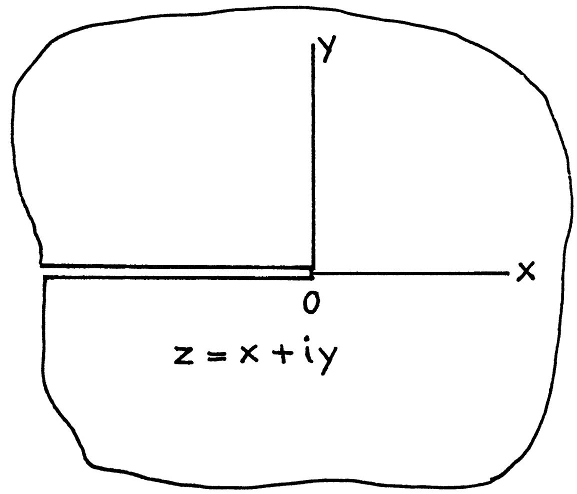
PART

III

TWO-DIMENSIONAL STRESS SOLUTIONS FOR VARIOUS CONFIGURATIONS WITH CRACKS

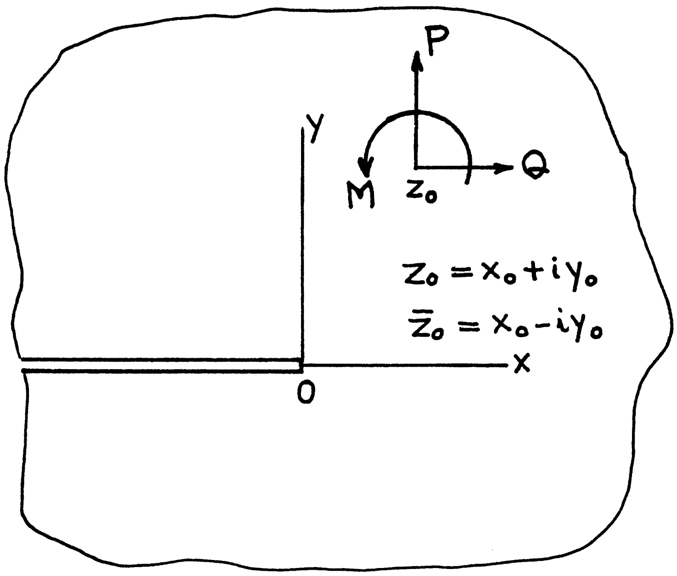


$$\begin{Bmatrix} Z_I(z) \\ Z_{II}(z) \\ Z_{III}(z) \end{Bmatrix} = \begin{Bmatrix} K_I \\ K_{II} \\ K_{III} \end{Bmatrix} \frac{1}{\sqrt{2\pi z}}$$

$$\begin{Bmatrix} \overline{Z}_I(z) \\ \overline{Z}_{II}(z) \\ \overline{Z}_{III}(z) \end{Bmatrix} = \frac{1}{\pi} \begin{Bmatrix} K_I \\ K_{II} \\ K_{III} \end{Bmatrix} \sqrt{2\pi z}$$

Method: Westergaard Stress Function
Accuracy: Exact
References: **Irwin 1958a** (see also **Williams 1957**)

NOTE: These Westergaard stress functions are the solutions for the crack-tip elastic field. That is, **Eqs. (1), (2), and (3)** in the text are directly derived from these functions by use of **Eqs. (38) and (39), Eqs. (55) and (56), and Eqs. (59) and (60),** respectively.



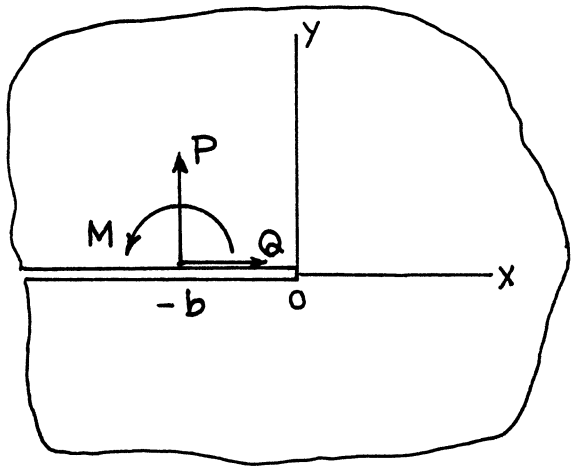
$$K = K_I - iK_{II}$$

$$= \frac{1}{\sqrt{2\pi}} \frac{1}{\kappa + 1} \left\{ (Q + iP) \left(\frac{1}{\sqrt{z_0}} - \kappa \frac{1}{\sqrt{\bar{z}_0}} \right) + \frac{(Q - iP)(\bar{z}_0 - z_0) + i(\kappa + 1)M}{2\,\bar{z}_0 \sqrt{\bar{z}_0}} \right\}$$

where

$$\kappa = \begin{cases} \frac{3-\nu}{1+\nu} & \text{plane stress} \\ 3 - 4\nu & \text{plane strain} \end{cases}$$

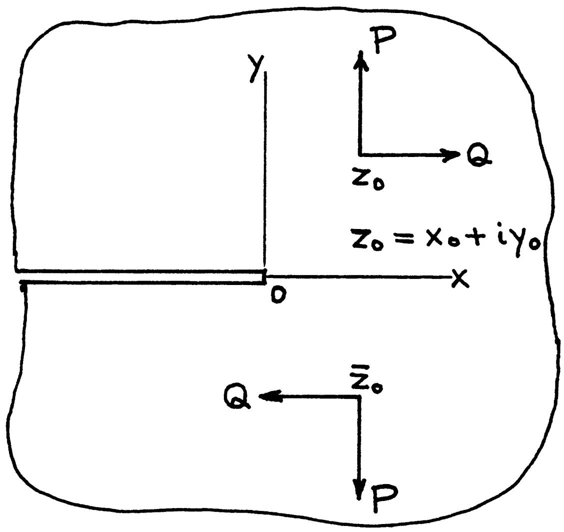
Method: Muskhelishvili's Method (Special Case of **page 5.3**)
Accuracy: Exact
References: **Erdogan 1962; Sih 1962a**



$$K_I = \frac{1}{\sqrt{2\pi b}}\left(P + \frac{M}{2b}\right)$$

$$K_{II} = \frac{1}{\sqrt{2\pi b}}Q$$

Method: Muskhelishvili's Method (Special Case of **page 5.4**)
Accuracy: Exact
Reference: **Erdogan 1962**



$$z = x + iy$$

$$\begin{Bmatrix} Z_I(z) \\ Z_{II}(z) \end{Bmatrix} = \frac{1}{\pi}\begin{Bmatrix} P \\ Q \end{Bmatrix}\left[1\begin{Bmatrix} - \\ + \end{Bmatrix}\alpha y_0 \frac{\partial}{\partial y_0}\right]\frac{1}{2}\left(\frac{1}{z-z_0}\sqrt{\frac{-z_0}{z}} + \frac{1}{z-\bar{z}_0}\sqrt{\frac{-\bar{z}_0}{z}}\right)$$

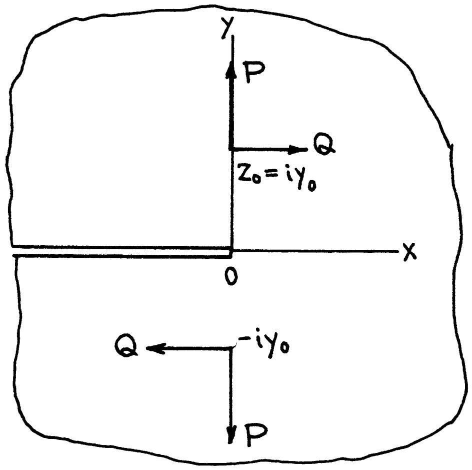
$$\begin{Bmatrix} \bar{Z}_I(z) \\ \bar{Z}_{II}(z) \end{Bmatrix} = \frac{1}{\pi}\begin{Bmatrix} P \\ Q \end{Bmatrix}\left[1\begin{Bmatrix} - \\ + \end{Bmatrix}\alpha y_0 \frac{\partial}{\partial y_0}\right]\left(\tan^{-1}\sqrt{\frac{z}{-z_0}} + \tan^{-1}\sqrt{\frac{z}{-\bar{z}_0}}\right)$$

$$\begin{Bmatrix} K_I \\ K_{II} \end{Bmatrix} = \frac{1}{\sqrt{2\pi}}\begin{Bmatrix} P \\ Q \end{Bmatrix}\left[1\begin{Bmatrix} - \\ + \end{Bmatrix}\alpha y_0 \frac{\partial}{\partial y_0}\right]\left\{\frac{1}{\sqrt{-z_0}} + \frac{1}{\sqrt{-\bar{z}_0}}\right\}$$

where

$$\alpha = \begin{cases} \frac{1}{2}(1+\nu) & \text{plane stress} \\ \frac{1}{2}\left(\frac{1}{1-\nu}\right) & \text{plane strain} \end{cases}$$

Method: Westergaard Stress Function
Accuracy: Exact
References: **Tada 1972a, 1973**



$z = x + iy$

$$\begin{Bmatrix} Z_I(z) \\ Z_{II}(z) \end{Bmatrix} = \frac{1}{\pi} \begin{Bmatrix} P \\ Q \end{Bmatrix} \left[1 \begin{Bmatrix} - \\ + \end{Bmatrix} \alpha y_0 \frac{\partial}{\partial y_0} \right] \frac{1}{\sqrt{2}} \frac{z + y_0}{z^2 + y_0^2} \sqrt{\frac{y_0}{z}}$$

$$\begin{Bmatrix} \overline{Z}_I(z) \\ \overline{Z}_{II}(z) \end{Bmatrix} = \frac{1}{\pi} \begin{Bmatrix} P \\ Q \end{Bmatrix} \left[1 \begin{Bmatrix} - \\ + \end{Bmatrix} \alpha y_0 \frac{\partial}{\partial y_0} \right] \frac{1}{2} \tan^{-1} \frac{\sqrt{2 y_0 z}}{y_0 - z}$$

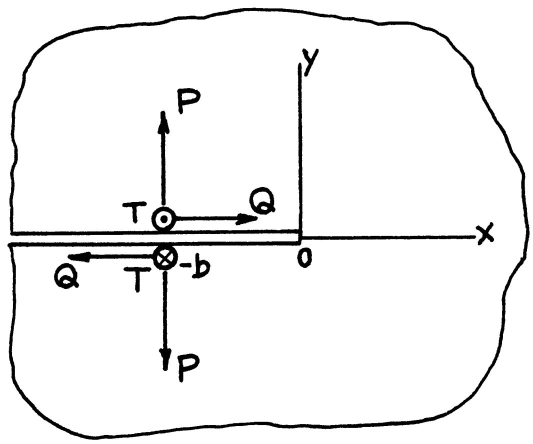
$$\begin{aligned} \begin{Bmatrix} K_I \\ K_{II} \end{Bmatrix} &= \frac{1}{\sqrt{\pi}} \begin{Bmatrix} P \\ Q \end{Bmatrix} \left[1 \begin{Bmatrix} - \\ + \end{Bmatrix} \alpha y_0 \frac{\partial}{\partial y_0} \right] \frac{1}{\sqrt{y_0}} \\ &= \begin{Bmatrix} P \\ Q \end{Bmatrix} \left[1 \begin{Bmatrix} + \\ - \end{Bmatrix} \frac{\alpha}{2} \right] \frac{1}{\sqrt{\pi y_0}} \end{aligned}$$

$$\begin{aligned} \operatorname{Im} \underset{x<0}{\begin{Bmatrix} \overline{Z}_I(x) \\ \overline{Z}_{II}(x) \end{Bmatrix}} &= \frac{1}{2\pi} \begin{Bmatrix} P \\ Q \end{Bmatrix} \left[1 \begin{Bmatrix} - \\ + \end{Bmatrix} \alpha y_0 \frac{\partial}{\partial y_0} \right] \tanh^{-1} \frac{\sqrt{2 y_0 |x|}}{y_0 - x} \\ &= \frac{1}{2\pi} \begin{Bmatrix} P \\ Q \end{Bmatrix} \left[\tanh^{-1} \frac{\sqrt{2 y_0 |x|}}{y_0 - x} \begin{Bmatrix} + \\ - \end{Bmatrix} \alpha \frac{x \sqrt{2 y_0 |x|}}{y_0^2 + x^2} \right] \end{aligned}$$

where

$$\alpha = \begin{cases} \frac{1}{2}(1 + \nu) & \text{plane stress} \\ \frac{1}{2}\left(\frac{1}{1-\nu}\right) & \text{plane strain} \end{cases}$$

Method: Westergaard Stress Function (Special Case of **page 3.4**)
Accuracy: Exact
References: **Tada 1972a, 1973**



$$z = x + iy$$

$$\begin{Bmatrix} Z_I(z) \\ Z_{II}(z) \\ Z_{III}(z) \end{Bmatrix} = \frac{1}{\pi} \begin{Bmatrix} P \\ Q \\ T \end{Bmatrix} \frac{1}{z+b} \sqrt{\frac{b}{z}}$$

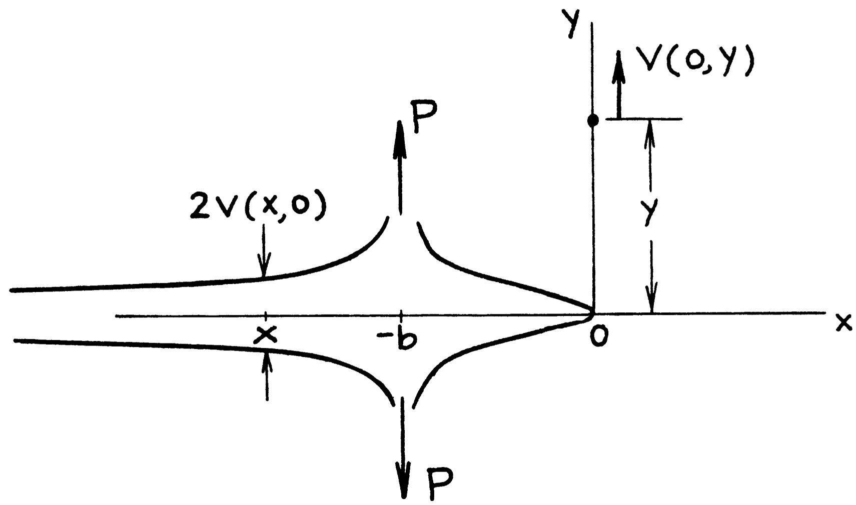
$$\begin{Bmatrix} \overline{Z}_I(z) \\ \overline{Z}_{II}(z) \\ \overline{Z}_{III}(z) \end{Bmatrix} = -\frac{2}{\pi} \begin{Bmatrix} P \\ Q \\ T \end{Bmatrix} \tan^{-1} \sqrt{\frac{b}{z}}$$

$$\begin{Bmatrix} K_I \\ K_{II} \\ K_{III} \end{Bmatrix} = \frac{\sqrt{2}}{\sqrt{\pi b}} \begin{Bmatrix} P \\ Q \\ T \end{Bmatrix}$$

$$\operatorname{Im} \begin{Bmatrix} \overline{Z}_I(x) \\ \overline{Z}_{II}(x) \\ \overline{Z}_{III}(x) \end{Bmatrix}_{x<0} = \frac{1}{\pi} \begin{Bmatrix} P \\ Q \\ T \end{Bmatrix} \ln \left| \frac{\sqrt{|x|}+\sqrt{b}}{\sqrt{|x|}-\sqrt{b}} \right|$$

$$\text{or } = \frac{2}{\pi} \begin{Bmatrix} P \\ Q \\ T \end{Bmatrix} \begin{cases} \tanh^{-1} \sqrt{|x|/b} & (-b < x \le 0) \\ \coth^{-1} \sqrt{|x|/b} & (x < -b) \end{cases}$$

Methods: Westergaard Stress Function, etc.
Accuracy: Exact
References: **Irwin 1957**, etc.



$$K_I = \frac{\sqrt{2}P}{\sqrt{\pi b}}$$

Crack Opening Profile

$$\underset{x \le 0}{2v(x,0)} = \frac{8P}{\pi E'} \begin{cases} \tanh^{-1}\sqrt{\frac{|x|}{b}} & -b < x \le 0 \\ \coth^{-1}\sqrt{\frac{|x|}{b}} & x < -b \end{cases}$$

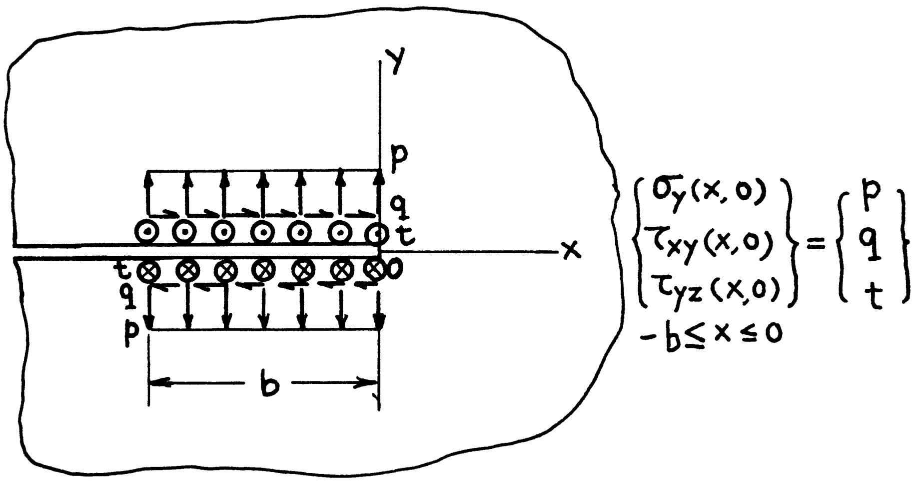
Vertical Displacement at $(0, y)$

$$v(0,y) = \frac{P}{\pi E'}\left[\tanh^{-1}\frac{\sqrt{2yb}}{y+b} - \alpha\frac{b\sqrt{2yb}}{y^2+b^2}\right]$$

where

$$\alpha = \begin{cases} \frac{1}{2}(1+\nu) & \text{plane stress} \\ \frac{1}{2}\left(\frac{1}{1-\nu}\right) & \text{plane strain} \end{cases}$$

Methods: $v(x,0)$ Westergaard Function (see **page 3.6**); $v(0,y)$ Paris' Equation (see **Appendix B**) or Reciprocity (see **page 3.5**)
Accuracy: Exact
Reference: **Tada 1985**



$$z = x + iy$$

$$\begin{Bmatrix} Z_I(z) \\ Z_{II}(z) \\ Z_{III}(z) \end{Bmatrix} = \frac{2}{\pi}\begin{Bmatrix} p \\ q \\ t \end{Bmatrix}\left\{\sqrt{\frac{b}{z}} - \tan^{-1}\sqrt{\frac{b}{z}}\right\}$$

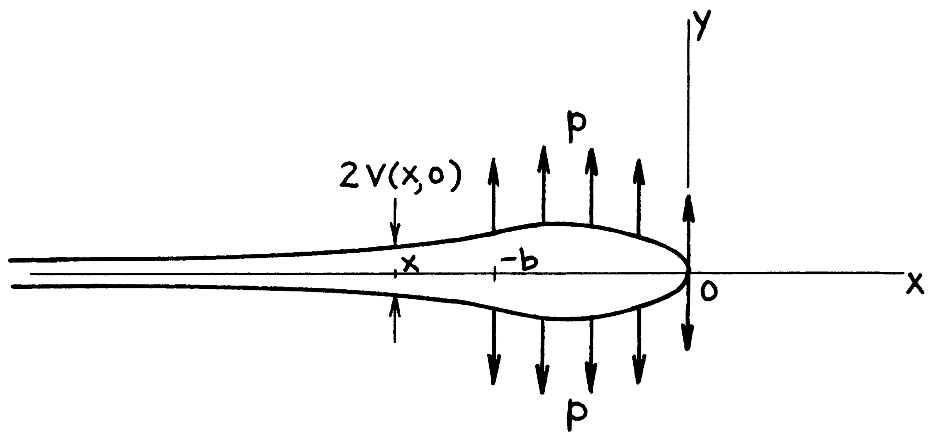
$$\begin{Bmatrix} \overline{Z}_I(z) \\ \overline{Z}_{II}(z) \\ \overline{Z}_{III}(z) \end{Bmatrix} = \frac{2}{\pi}\begin{Bmatrix} p \\ q \\ t \end{Bmatrix} b\left\{\sqrt{\frac{z}{b}} - \left(1 + \frac{z}{b}\right)\tan^{-1}\sqrt{\frac{b}{z}}\right\}$$

$$\begin{Bmatrix} K_I \\ K_{II} \\ K_{III} \end{Bmatrix} = \frac{2}{\pi}\begin{Bmatrix} p \\ q \\ t \end{Bmatrix}\sqrt{2\pi b}$$

$$\operatorname{Im}\underset{x<0}{\begin{Bmatrix} \overline{Z}_I(x) \\ \overline{Z}_{II}(x) \\ \overline{Z}_{III}(x) \end{Bmatrix}} = \frac{2}{\pi}\begin{Bmatrix} p \\ q \\ t \end{Bmatrix} b\left\{\sqrt{\frac{|x|}{b}} + \left(1 + \frac{x}{b}\right)\frac{1}{2}\ln\left|\frac{\sqrt{|x|} + \sqrt{b}}{\sqrt{|x|} - \sqrt{b}}\right|\right\}$$

$$\operatorname{Im}\underset{x=-b}{\begin{Bmatrix} \overline{Z}_I(x) \\ \overline{Z}_{II}(x) \\ \overline{Z}_{III}(x) \end{Bmatrix}} = \frac{2}{\pi}\begin{Bmatrix} p \\ q \\ t \end{Bmatrix} b$$

Methods: Westergaard Stress Function, etc. (Integration of **page 3.6**)
Accuracy: Exact
Reference: **Tada 1973**



$$K_I = \frac{2\sqrt{2}}{\sqrt{\pi}} p\sqrt{b}$$

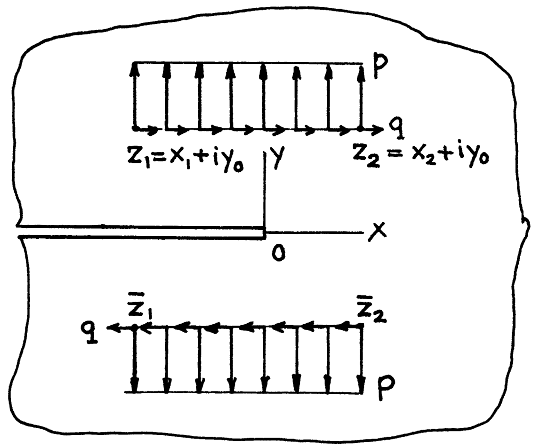
Crack Opening Profile

$$\underset{x \le 0}{2v(x,0)} = \frac{8p}{\pi E'} b \left[\sqrt{\frac{|x|}{b}} + \left(1 + \frac{x}{b}\right) \left\{ \begin{array}{ll} \tanh^{-1} \sqrt{\frac{|x|}{b}} & -b < x \le 0 \\ \coth^{-1} \sqrt{\frac{|x|}{b}} & x < -b \end{array} \right\} \right]$$

Opening at $(-b, 0)$:

$$2v(-b,0) = \frac{8p}{\pi E'} b$$

Method: Westergaard Stress Function (see **page 3.7**.)
Accuracy: Exact
Reference: **Tada 1985**



$$z = x + iy$$

$$\begin{Bmatrix} Z_I(z) \\ Z_{II}(z) \end{Bmatrix} = \frac{1}{\pi}\begin{Bmatrix} p \\ q \end{Bmatrix}\left[1\begin{Bmatrix} - \\ + \end{Bmatrix}\alpha y_0\frac{\partial}{\partial y_0}\right]\left[\left\{\left(\sqrt{-z_2}-\sqrt{-\bar{z}_2}\right)-\left(\sqrt{-z_1}-\sqrt{-\bar{z}_1}\right)\right\}\frac{1}{\sqrt{z}}\right.$$
$$\left.-\left\{\left(\tan^{-1}\sqrt{\frac{-z_2}{z}}-\tan^{-1}\sqrt{\frac{-\bar{z}_2}{z}}\right)-\left(\tan^{-1}\sqrt{\frac{-z_1}{z}}-\tan^{-1}\sqrt{\frac{-\bar{z}_1}{z}}\right)\right\}\right]$$

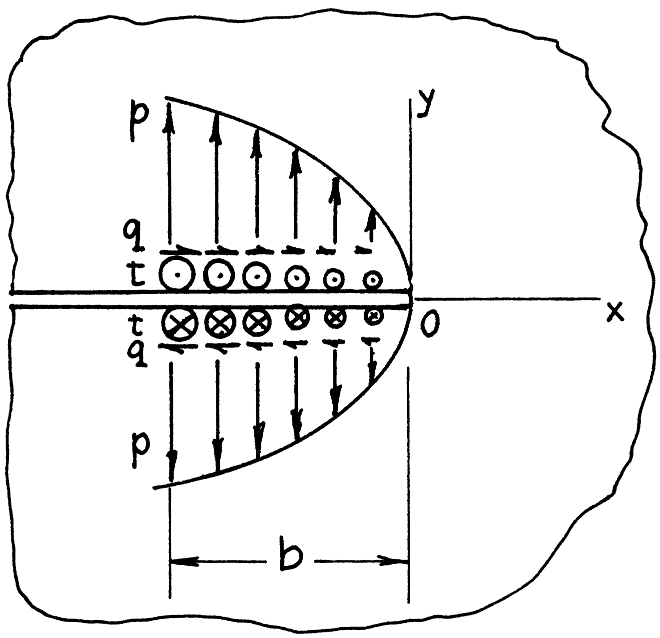
$$\begin{Bmatrix} \overline{Z}_I(z) \\ \overline{Z}_{II}(z) \end{Bmatrix} = \frac{1}{\pi}\begin{Bmatrix} p \\ q \end{Bmatrix}\left[1\begin{Bmatrix} - \\ + \end{Bmatrix}\alpha y_0\frac{\partial}{\partial y_0}\right]\left[\left\{\left(\sqrt{-z_2}-\sqrt{-\bar{z}_2}\right)-\left(\sqrt{-z_1}-\sqrt{-\bar{z}_1}\right)\right\}\sqrt{z}\right.$$
$$\left.-\left\{(z-z_2)\tan^{-1}\sqrt{\frac{-z_2}{z}}-(z-\bar{z}_2)\tan^{-1}\sqrt{\frac{-\bar{z}_2}{z}}-(z-z_1)\tan^{-1}\sqrt{\frac{-z_1}{z}}+(z-\bar{z}_1)\tan^{-1}\sqrt{\frac{-\bar{z}_1}{z}}\right\}\right]$$

$$\begin{Bmatrix} K_I \\ K_{II} \end{Bmatrix} = \frac{2}{\sqrt{2\pi}}\begin{Bmatrix} p \\ q \end{Bmatrix}\left[1\begin{Bmatrix} - \\ + \end{Bmatrix}\alpha y_0\frac{\partial}{\partial y_0}\right]\left\{\left(\sqrt{-z_2}-\sqrt{-\bar{z}_2}\right)-\left(\sqrt{-z_1}-\sqrt{-\bar{z}_1}\right)\right\}$$

where

$$\alpha = \begin{cases} \frac{1}{2}(1+\nu) & \text{plane stress} \\ \frac{1}{2}\left(\frac{1}{1-\nu}\right) & \text{plane strain} \end{cases}$$

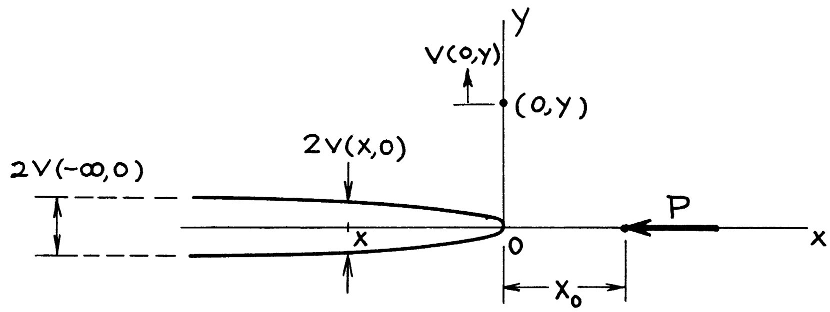
Method: Westergaard Stress Function (Integration of **page 3.4**)
Accuracy: Exact
References: **Tada 1972a, 1973**



$$\left\{\begin{array}{c}\sigma_y(x,0)\\ \tau_{xy}(x,0)\\ \tau_{yz}(x,0)\end{array}\right\}_{-b\le x\le 0} = \left\{\begin{array}{c}p\\ q\\ t\end{array}\right\}\left(\frac{-x}{b}\right)^{\gamma} \quad \left(\gamma > -{}^{1}\!/_{2}\right)$$

$$\left\{\begin{array}{c}K_I\\ K_{II}\\ K_{III}\end{array}\right\} = \frac{2}{\pi}\left\{\begin{array}{c}p\\ q\\ t\end{array}\right\}\sqrt{2\pi b}\ \frac{1}{2\gamma+1}$$

Method: Integration of **page 3.6**
Accuracy: Exact
Reference: **Tada 1972a, 1974**



$$z = x + iy$$

$$Z_I(z) = \frac{P}{2\pi}(1-\alpha)\frac{\sqrt{x_0}}{(x_0 - z)\sqrt{z}}$$

$$\overline{Z}_I(z) = \frac{P}{\pi}(1-\alpha)\tanh^{-1}\sqrt{\frac{z}{x_0}}\left(= \frac{P}{2\pi}(1-\alpha)\cosh^{-1}\frac{z+x_0}{z-x_0}\right)$$

$$K_I = \frac{P}{\sqrt{2\pi x_0}}(1-\alpha)$$

Crack Opening Profile:

$$\underset{x\leq 0}{2v(x,0)} = \frac{4P}{\pi E'}(1-\alpha)\tan^{-1}\sqrt{\frac{|x|}{x_0}}$$

Opening at Infinity:

$$2v(-\infty, 0) = \frac{2P}{E'}(1-\alpha)$$

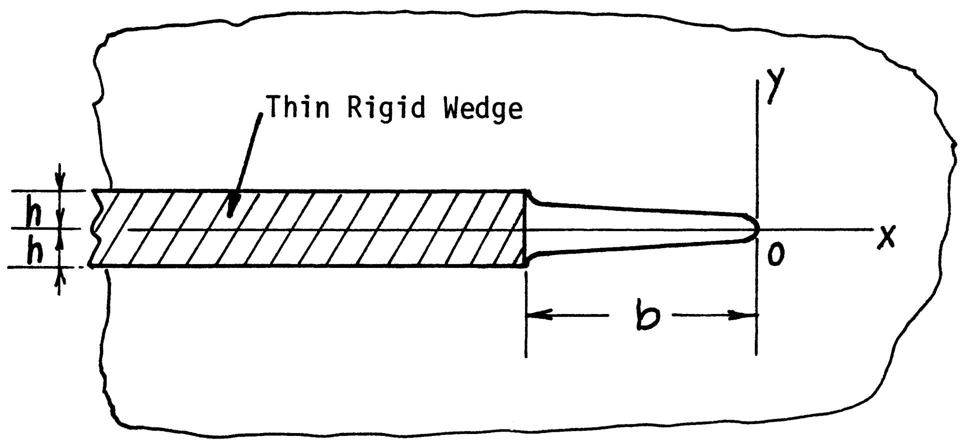
Vertical Displacement at $(0, y)$:

$$v(0,y) = \frac{P}{\pi E'}(1-\alpha)\left\{\sin^{-1}\sqrt{\frac{2x_0 y}{x_0^2 + y^2}} - \frac{\alpha}{2}\frac{\sqrt{2x_0 y}(x_0 + y)}{x_0^2 + y^2}\right\}$$

where

$$\alpha = \begin{cases} \frac{1}{2}(1+\nu) & \text{Plane Stress} \\ \frac{1}{2}\left(\frac{1}{1-\nu}\right) & \text{Plane Strain} \end{cases}$$

Method: Integration of **pages 3.6, 3.6a** or Special Case of **page 5.19 or 5.20, or 24.19**
Accuracy: Exact
Reference: **Tada 1985**



$$\underset{x=0}{K_I} = \frac{E'h}{\sqrt{2\pi b}}$$

$$\underset{x=-b}{K_I} = -\frac{E'h}{\sqrt{2\pi b}}$$

$$\underset{-b<x<0}{2v(x,0)} = \frac{4}{\pi} h \sin^{-1} \sqrt{\frac{-x}{b}}$$

where

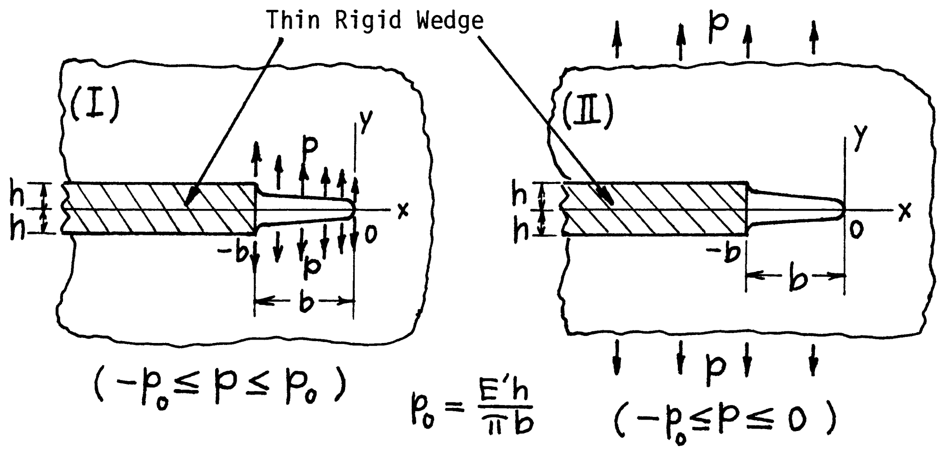
$$E' = \begin{cases} E & \text{plane stress} \\ E/(1-\nu^2) & \text{plane strain} \end{cases}$$

Methods: Singular Integral Equation, Westergaard Stress Function or a Special Case of **page 4.15** or **page 5.21**
Accuracy: Exact
References: **Barenblatt 1962, Tada 1985**

The Westergaard Stress Functions are

$$Z_I(z) = \frac{E'h}{2\pi} \cdot \frac{1}{\sqrt{z(z+b)}}$$

$$\overline{Z}_I(z) = \frac{E'h}{2\pi} \sinh^{-1} \sqrt{\frac{z}{b}}$$



$$K_I\underset{x=0}{} = \frac{E'h}{\sqrt{2\pi b}} + p\sqrt{\pi \cdot b/2}$$

$$K_I\underset{x=-b}{} = -\frac{E'h}{\sqrt{2\pi b}} + p\sqrt{\pi \cdot b/2}$$

$$\underset{-b \leq x \leq 0}{2v(x,0)} = \frac{4}{\pi} h \sin^{-1}\sqrt{\frac{-x}{b}} + \frac{4p}{E'}\sqrt{-x(b-x)}$$

where

$$E' = \begin{cases} E & \text{plane stress} \\ E/(1-\nu^2) & \text{plane strain} \end{cases}$$

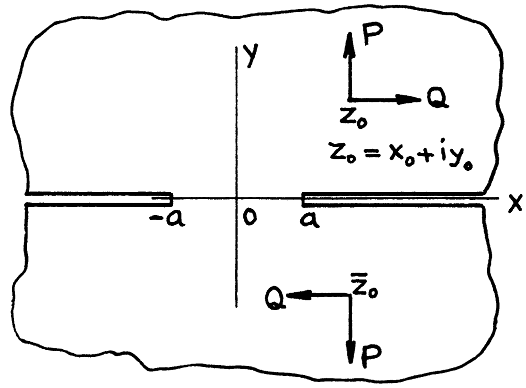
Method: Superposition of Solutions of **page 3.11** and **page 5.1**
Accuracy: Exact
References: **Tada 1974**

NOTE: 1. In (I), when $p > p_0$, separation of contact surfaces occurs near $x = -b$, and when $p < -p_0$, crack closure occurs near $x = 0$.
2. In (II), when $p > 0$ (remote tension), separation of contact surfaces occurs for large $-x$, and when $p < -p_0$, crack closure occurs near $x = 0$.

The Westergaard Functions are

$$\left\{ \begin{matrix} Z_I(z) \\ \overline{Z}_I(z) \end{matrix} \right\} = \left\{ \begin{matrix} Z_I(z) \\ \overline{Z}_I(z) \end{matrix} \right\} + \left\{ \begin{matrix} Z_I(z) \\ \overline{Z}_I(z) \end{matrix} \right\}$$

p. 3.12 **p. 3.11** **p. 5.1** (replace σ by p)



$$z = x + iy$$

$$\begin{Bmatrix} Z_I(z) \\ Z_{II}(z) \end{Bmatrix} = \frac{1}{\pi}\begin{Bmatrix} P \\ Q \end{Bmatrix}\left[1\begin{Bmatrix} - \\ + \end{Bmatrix}\alpha y_0 \frac{\partial}{\partial y_0}\right]\frac{1}{\sqrt{a^2 - z^2}}\left[\frac{1}{2}\left(\frac{\sqrt{z_0^2 - a^2}}{z_0 - z} - \frac{\sqrt{\bar{z}_0^2 - a^2}}{\bar{z}_0 - z}\right)\right.$$
$$\left. + \begin{Bmatrix} 1 \\ 0 \end{Bmatrix}\frac{z}{a^2}\left(\sqrt{z_0^2 - a^2} + \sqrt{\bar{z}_0^2 - a^2}\right)\right]$$

$$\begin{Bmatrix} \overline{Z}_I(z) \\ \overline{Z}_{II}(z) \end{Bmatrix} = \frac{1}{\pi}\begin{Bmatrix} P \\ Q \end{Bmatrix}\left[1\begin{Bmatrix} - \\ + \end{Bmatrix}\alpha y_0 \frac{\partial}{\partial y_0}\right]\left[\frac{1}{2}\left(\sin^{-1}\frac{z_0 z - a^2}{a(z_0 - z)} + \sin^{-1}\frac{\bar{z}_0 z - a^2}{a(\bar{z}_0 - z)}\right)\right.$$
$$\left. - \begin{Bmatrix} 1 \\ 0 \end{Bmatrix}\frac{\sqrt{a^2 - z^2}}{a^2}\left(\sqrt{z_0^2 - a^2} + \sqrt{\bar{z}_0^2 - a^2}\right)\right]$$

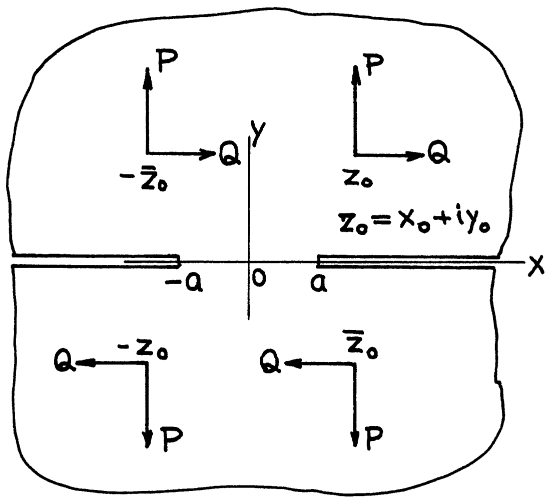
$$K_{I\pm a} = \frac{P}{\sqrt{\pi a}}\left[1 - \alpha y_0 \frac{\partial}{\partial y_0}\right]\left[\frac{1}{2}\left(\sqrt{\frac{z_0 \pm a}{z_0 \mp a}} + \sqrt{\frac{\bar{z}_0 \pm a}{\bar{z}_0 \mp a}}\right) \pm \frac{1}{a}\left(\sqrt{z_0^2 - a^2} + \sqrt{\bar{z}_0^2 - a^2}\right)\right]$$

$$K_{II\pm a} = \frac{Q}{\sqrt{\pi a}}\left[1 + \alpha y_0 \frac{\partial}{\partial y_0}\right]\frac{1}{2}\left(\sqrt{\frac{z_0 \pm a}{z_0 \mp a}} + \sqrt{\frac{\bar{z}_0 \pm a}{\bar{z}_0 \mp a}}\right)$$

where

$$\alpha = \begin{cases} \frac{1}{2}(1 + \nu) & \text{plane stress} \\ \frac{1}{2}\left(\frac{1}{1-\nu}\right) & \text{plane strain} \end{cases}$$

Method: Westergaard Stress Function
Accuracy: Exact
Reference: **Tada 2000**



$$z = x + iy$$

$$\begin{Bmatrix} Z_I(z) \\ Z_{II}(z) \end{Bmatrix} = \frac{1}{\pi} \begin{Bmatrix} P \\ Q \end{Bmatrix} \left[1 \begin{Bmatrix} - \\ + \end{Bmatrix} \alpha y_0 \frac{\partial}{\partial y_0} \right] \left(\frac{z_0}{z_0^2 - z^2} \sqrt{z_0^2 - a^2} + \frac{\bar{z}_0}{\bar{z}_0^2 - z^2} \sqrt{\bar{z}_0^2 - a^2} \right) \frac{1}{\sqrt{a^2 - z^2}}$$

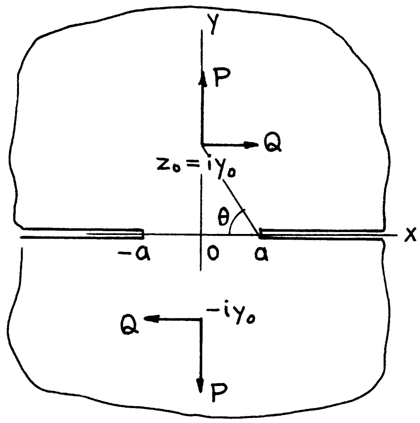
$$\begin{Bmatrix} \overline{Z}_I(z) \\ \overline{Z}_{II}(z) \end{Bmatrix} = \frac{1}{\pi} \begin{Bmatrix} P \\ Q \end{Bmatrix} \left[1 \begin{Bmatrix} - \\ + \end{Bmatrix} \alpha y_0 \frac{\partial}{\partial y_0} \right] \left\{ \tan^{-1} \sqrt{\frac{(a/z)^2 - 1}{1 - (a/z_0)^2}} + \tan^{-1} \sqrt{\frac{(a/z)^2 - 1}{1 - (a/\bar{z}_0)^2}} \right\}$$

$$\begin{Bmatrix} K_I \\ K_{II} \end{Bmatrix} = \frac{1}{\sqrt{\pi a}} \begin{Bmatrix} P \\ Q \end{Bmatrix} \left[1 \begin{Bmatrix} - \\ + \end{Bmatrix} \alpha y_0 \frac{\partial}{\partial y_0} \right] \left\{ \frac{z_0}{\sqrt{z_0^2 - a^2}} + \frac{\bar{z}_0}{\sqrt{\bar{z}_0^2 - a^2}} \right\}$$

where

$$\alpha = \begin{cases} \frac{1}{2}(1 + \nu) & \text{plane stress} \\ \frac{1}{2}\left(\frac{1}{1-\nu}\right) & \text{plane strain} \end{cases}$$

Method: Westergaard Stress Function (Superposition of **page 4.1**)
Accuracy: Exact
References: **Tada 1972a, 1973**



$z = x + iy$

$$\left\{\begin{matrix} Z_I(z) \\ Z_{II}(z) \end{matrix}\right\} = \frac{1}{\pi}\left\{\begin{matrix} P \\ Q \end{matrix}\right\}\left[1\left\{\begin{matrix} - \\ + \end{matrix}\right\}\alpha y_0 \frac{\partial}{\partial y_0}\right]\frac{y_0}{z^2 + y_0^2}\sqrt{\frac{a^2 + y_0^2}{a^2 - z^2}}$$

$$\left\{\begin{matrix} \overline{Z}_I(z) \\ \overline{Z}_{II}(z) \end{matrix}\right\} = \frac{1}{\pi}\left\{\begin{matrix} P \\ Q \end{matrix}\right\}\left[1\left\{\begin{matrix} - \\ + \end{matrix}\right\}\alpha y_0 \frac{\partial}{\partial y_0}\right]\tan^{-1}\sqrt{\frac{\left(a/z\right)^2 - 1}{1 + \left(a/y_0\right)^2}}$$

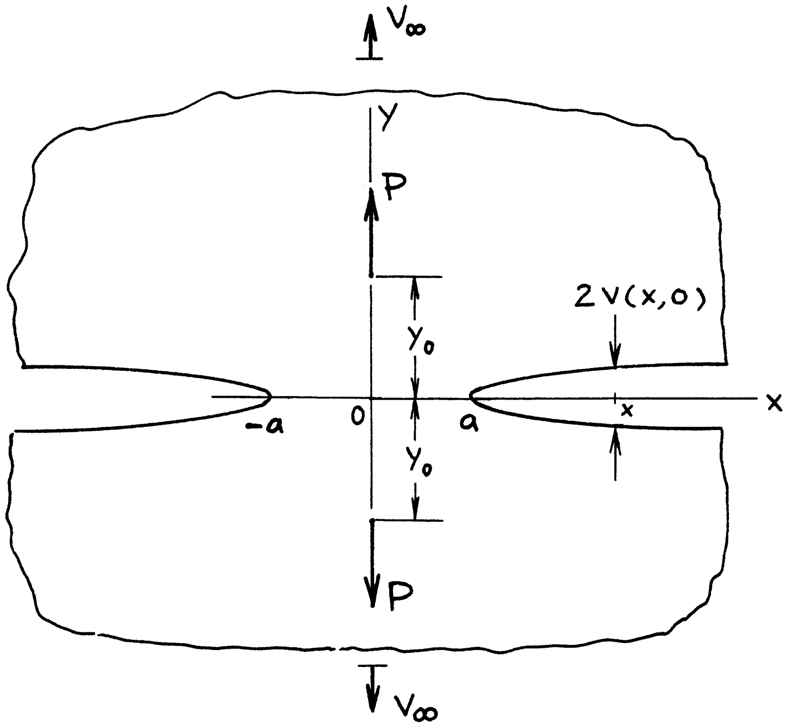
$$\begin{aligned}\left\{\begin{matrix} K_I \\ K_{II} \end{matrix}\right\} &= \frac{1}{\sqrt{\pi a}}\left\{\begin{matrix} P \\ Q \end{matrix}\right\}\left[1\left\{\begin{matrix} - \\ + \end{matrix}\right\}\alpha y_0 \frac{\partial}{\partial y_0}\right]\frac{y_0}{\sqrt{a^2 + y_0^2}} \\ &= \frac{1}{\sqrt{\pi a}}\left\{\begin{matrix} P \\ Q \end{matrix}\right\}\frac{y_0}{\sqrt{a^2 + y_0^2}}\left[1\left\{\begin{matrix} - \\ + \end{matrix}\right\}\alpha\frac{a^2}{a^2 + y_0^2}\right]\left(= \frac{1}{\sqrt{\pi a}}\left\{\begin{matrix} P \\ Q \end{matrix}\right\}\sin\theta\left[1\left\{\begin{matrix} - \\ + \end{matrix}\right\}\alpha\cos^2\theta\right]\right)\end{aligned}$$

$$\begin{aligned}\underset{|x| \geq a}{\mathrm{Im}\left\{\begin{matrix} \overline{Z}_I(x) \\ \overline{Z}_{II}(x) \end{matrix}\right\}} &= \frac{1}{\pi}\left\{\begin{matrix} P \\ Q \end{matrix}\right\}\left[1\left\{\begin{matrix} - \\ + \end{matrix}\right\}\alpha y_0 \frac{\partial}{\partial y_0}\right]\tanh^{-1}\sqrt{\frac{1 - \left(a/x\right)^2}{1 + \left(a/y_0\right)^2}} \\ &= \frac{1}{\pi}\left\{\begin{matrix} P \\ Q \end{matrix}\right\}\left[\tanh^{-1}\sqrt{\frac{1 - \left(a/x\right)^2}{1 + \left(a/y_0\right)^2}}\left\{\begin{matrix} - \\ + \end{matrix}\right\}\alpha\frac{y_0 x}{x^2 + y_0^2}\sqrt{\frac{x^2 - a^2}{a^2 + y_0^2}}\right]\end{aligned}$$

where

$$\alpha = \begin{cases} \frac{1}{2}(1 + \nu) & \text{plane stress} \\ \frac{1}{2}\left(\frac{1}{1-\nu}\right) & \text{plane strain} \end{cases}$$

Method: Westergaard Stress Function (Special Case of **page 4.1**)
Accuracy: Exact
References: **Tada 1972a, 1973**



$$K_I = \frac{P}{\sqrt{\pi a}} \frac{y_0}{\sqrt{a^2 + y_0^2}} \left(1 - \alpha \frac{y_0}{\sqrt{a^2 + y_0^2}} \right)$$

Crack Opening Profile:

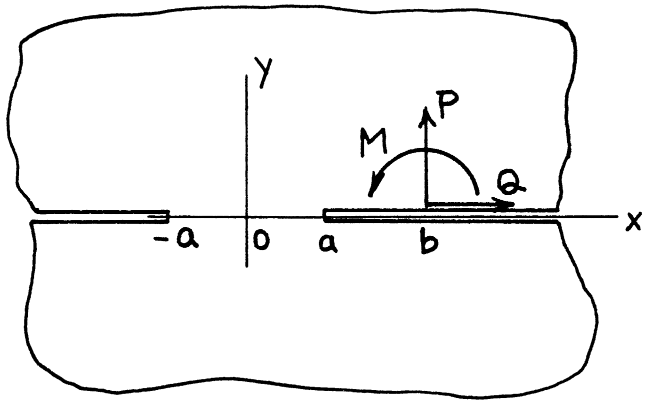
$$\underset{|x| \geq a}{2v(x, 0)} = \frac{4P}{\pi E'} \left(\tanh^{-1} \sqrt{\frac{1 - (a/x)^2}{1 + (a/y_0)^2}} - \alpha \frac{x^2}{x^2 + y_0^2} \sqrt{\frac{1 - (a/x)^2}{1 + (a/y_0)^2}} \right)$$

Relative Vertical Displacement at Infinity:

$$2v_\infty = \frac{4P}{\pi E'} \left(\sinh^{-1} \frac{y_0}{a} - \alpha \frac{y_0}{\sqrt{a^2 + y_0^2}} \right)$$

$$2v_\infty = 2v(x, 0)_{|x| \to \infty}$$

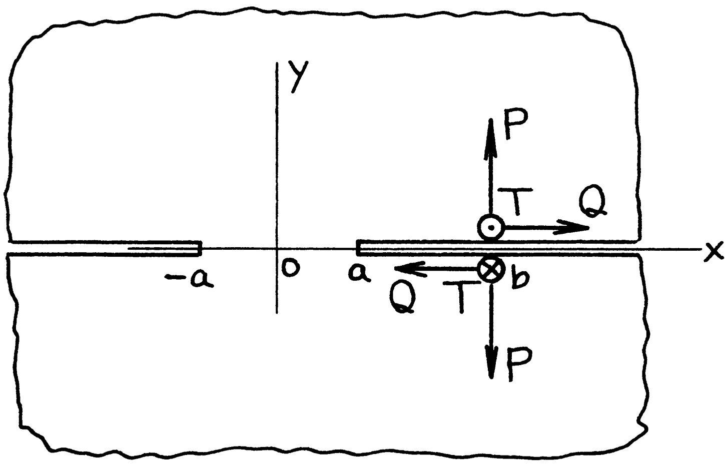
Methods: $v(x, 0)$ Westergaard Stress Function (see **page 4.3**); v_∞ Paris' Equation (see **Appendix B**) or Reciprocity (see **page 4.9**)
Accuracy: Exact
Reference: **Tada 1985**



$$K_{I\pm a} = \frac{P}{2\sqrt{\pi a}}\sqrt{\frac{b\pm a}{b\mp a}} \mp \frac{M}{2\sqrt{\pi a}}\frac{a}{(b\mp a)\sqrt{b^2 - a^2}}$$

$$K_{II\pm a} = \frac{Q}{2\sqrt{\pi a}}\sqrt{\frac{b\pm a}{b\mp a}}$$

Method: Muskhelishvili's Method (Special Case of **page 6.2**)
Accuracy: Exact
Reference: **Erdogan 1962**



$$z = x + iy$$

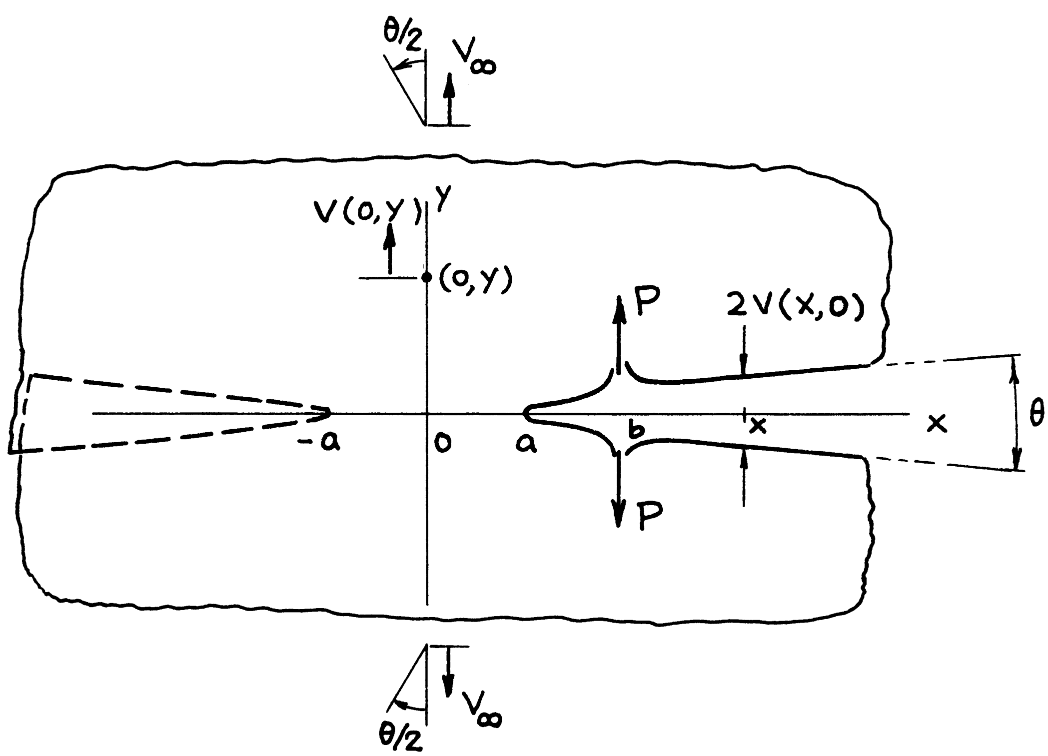
$$\begin{Bmatrix} Z_I(z) \\ Z_{II}(z) \\ Z_{III}(z) \end{Bmatrix} = \frac{1}{\pi}\begin{Bmatrix} P \\ Q \\ T \end{Bmatrix} \frac{\sqrt{b^2 - a^2}}{\sqrt{a^2 - z^2}} \left(\frac{1}{b - z} + \frac{2}{a^2} z \begin{Bmatrix} 1 \\ 0 \\ 1 \end{Bmatrix} \right)$$

$$\begin{Bmatrix} \overline{Z}_I(z) \\ \overline{Z}_{II}(z) \\ \overline{Z}_{III}(z) \end{Bmatrix} = \frac{1}{\pi}\begin{Bmatrix} P \\ Q \\ T \end{Bmatrix} \left[\sin^{-1} \frac{bz - a^2}{a(b - z)} - \frac{2}{a^2} \sqrt{b^2 - a^2} \sqrt{a^2 - z^2} \begin{Bmatrix} 1 \\ 0 \\ 1 \end{Bmatrix} \right]$$

$$\begin{Bmatrix} K_I \\ K_{II} \\ K_{III} \end{Bmatrix}_{\pm a} = \frac{1}{\sqrt{\pi a}}\begin{Bmatrix} P \\ Q \\ T \end{Bmatrix} \sqrt{b^2 - a^2} \left(\frac{1}{b \mp a} \pm \frac{2}{a} \begin{Bmatrix} 1 \\ 0 \\ 1 \end{Bmatrix} \right)$$

$$\underset{|x| \geq a}{\mathrm{Im}\begin{Bmatrix} \overline{Z}_I(x) \\ \overline{Z}_{II}(x) \\ \overline{Z}_{III}(x) \end{Bmatrix}} = \frac{1}{\pi}\begin{Bmatrix} P \\ Q \\ T \end{Bmatrix} \left[\cosh^{-1} \left| \frac{bx - a^2}{a(b - x)} \right| + 2\frac{x}{a} \sqrt{\left(\frac{b}{a}\right)^2 - 1} \sqrt{1 - \left(\frac{a}{x}\right)^2} \begin{Bmatrix} 1 \\ 0 \\ 1 \end{Bmatrix} \right]$$

Method: Westergaard Stress Function
Accuracy: Exact
Reference: **Tada 1985**



$$K_{I\pm a} = \frac{P}{\sqrt{\pi a}}\sqrt{b^2 - a^2}\left(\frac{1}{b \mp a} \pm \frac{2}{a}\right)$$

Crack Opening Profile:

$$\underset{|x|\geq a}{2v(x,0)} = \frac{4P}{\pi E'}\left[\cosh^{-1}\left|\frac{bx - a^2}{a(b-x)}\right| + \frac{x}{a}\sqrt{\left(\frac{b}{a}\right)^2 - 1}\sqrt{1 - \left(\frac{a}{x}\right)^2}\right]$$

Vertical Displacement at $(0, y)$:

$$v(0,y) = \frac{2P}{\pi E'}\left[\tanh^{-1}\sqrt{\frac{1-(a/b)^2}{1+(a/y)^2}} - \alpha\frac{b^2}{b^2+y^2}\sqrt{\frac{1-(a/b)^2}{1+(a/y)^2}}\right]$$

Relative Vertical Displacement along y-Axis at Infinity:

$$2v_\infty = \frac{4P}{\pi E'}\cosh^{-1}\frac{b}{a}$$

Relative Rotation at Infinity:

$$\theta = \frac{8P}{\pi E'} \frac{\sqrt{b^2 - a^2}}{a^2} \left(= \left\{ \frac{2v(x,0)}{x} \right\}_{x\to\infty} \right)$$

where

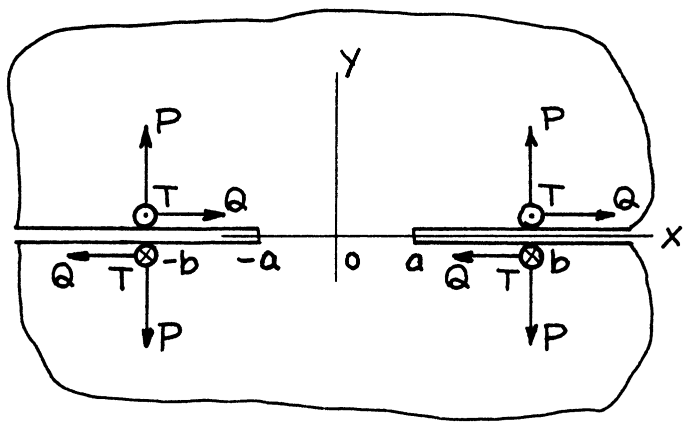
$$\alpha = \begin{cases} \frac{1}{2}(1+\nu) & \text{plane stress} \\ \frac{1}{2}\left(\frac{1}{1-\nu}\right) & \text{plane strain} \end{cases}$$

Methods: v Reciprocity (see **page 4.3a**); θ From **page 4.10a** with $M = P\sqrt{b^2 - a^2}$ (or Paris' Equation — see **Appendix B**)

Accuracy: Exact

Reference: **Tada 1985**

NOTE: 1. Always $K_{I-a} < 0$.
2. No surface interference $(x \leq -a)$ was considered.



$$z = x + iy$$

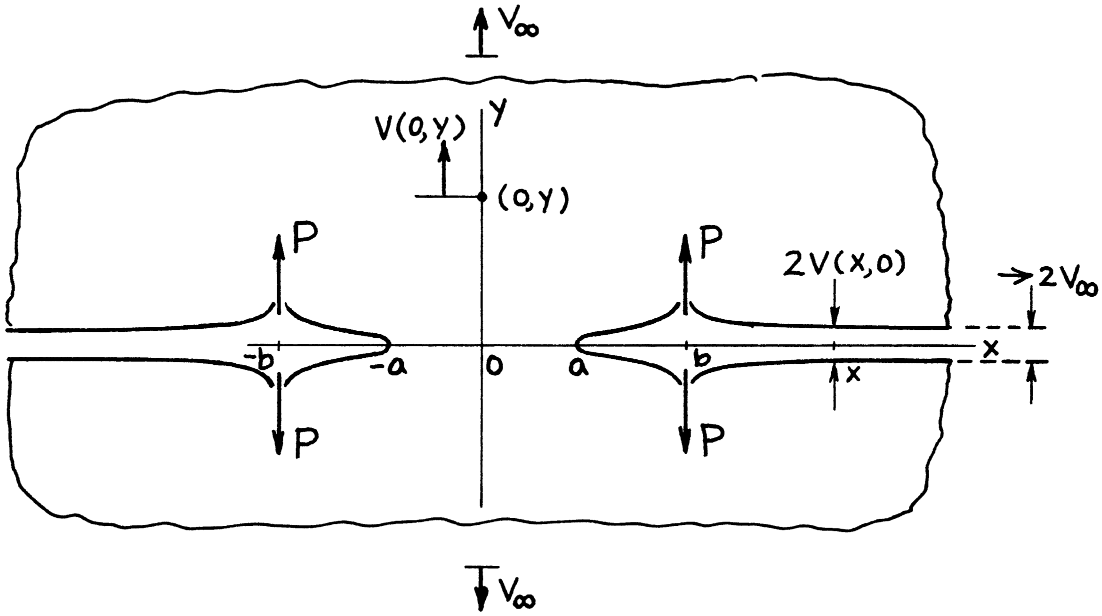
$$\begin{Bmatrix} Z_I(z) \\ Z_{II}(z) \\ Z_{III}(z) \end{Bmatrix} = \frac{2}{\pi} \begin{Bmatrix} P \\ Q \\ T \end{Bmatrix} \frac{b\sqrt{b^2 - a^2}}{\left(b^2 - z^2\right)\sqrt{a^2 - z^2}}$$

$$\begin{Bmatrix} \overline{Z}_I(z) \\ \overline{Z}_{II}(z) \\ \overline{Z}_{III}(z) \end{Bmatrix} = \frac{2}{\pi} \begin{Bmatrix} P \\ Q \\ T \end{Bmatrix} \tan^{-1} \sqrt{\frac{1 - (a/b)^2}{(a/z)^2 - 1}}$$

$$\begin{Bmatrix} K_I \\ K_{II} \\ K_{III} \end{Bmatrix} = \frac{2}{\sqrt{\pi a}} \begin{Bmatrix} P \\ Q \\ T \end{Bmatrix} \frac{b}{\sqrt{b^2 - a^2}}$$

$$\operatorname{Im} \underset{|x| \geq a}{\begin{Bmatrix} \overline{Z}_I(x) \\ \overline{Z}_{II}(x) \\ \overline{Z}_{III}(x) \end{Bmatrix}} = \frac{1}{\pi} \begin{Bmatrix} P \\ Q \\ T \end{Bmatrix} \ln \left| \frac{\sqrt{1 - (a/b)^2} + \sqrt{1 - (a/x)^2}}{\sqrt{1 - (a/b)^2} - \sqrt{1 - (a/x)^2}} \right|$$

Methods: Superposition of **page 4.5** (or Special Case of Periodic Cracks, **page 7.7**)
Accuracy: Exact
References: **Erdogan 1962; Tada 1973**



$$K_I = \frac{2P}{\sqrt{\pi a}} \, \frac{b}{\sqrt{b^2 - a^2}}$$

Crack Opening Profile:

$$\underset{|x| \geq a}{2v(x,0)} = \frac{8P}{\pi E'} \begin{Bmatrix} \tanh^{-1} \\ \coth^{-1} \end{Bmatrix} \sqrt{\frac{1 - (a/x)^2}{1 - (a/b)^2}} \quad \begin{matrix} a \leq |x| < b \\ |x| > b \end{matrix}$$

Vertical Displacement at $(0, y)$:

$$v(0,y) = \frac{4P}{\pi E'} \left[\tanh^{-1} \sqrt{\frac{1 - (a/b)^2}{1 + (a/y)^2}} - \alpha \frac{b^2}{b^2 + y^2} \sqrt{\frac{1 - (a/b)^2}{1 + (a/y)^2}} \right]$$

Relative Vertical Displacement at Infinity:

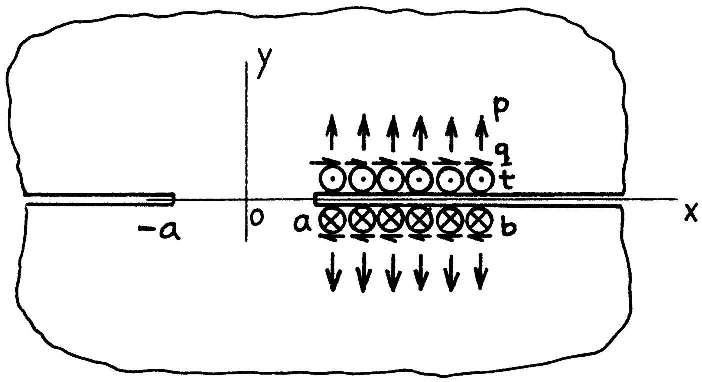
$$2v_\infty = \frac{8P}{\pi E'} \cosh^{-1} \frac{b}{a}$$

$$2v_\infty = \underset{y \to \infty}{2v(0,y)} = \underset{x \to \infty}{2v(x,0)}$$

where

$$\alpha = \begin{cases} \frac{1}{2}(1+\nu) & \text{plane stress} \\ \frac{1}{2}\left(\frac{1}{1-\nu}\right) & \text{plane strain} \end{cases}$$

Method: v Paris' Equation (see **Appendix B**) (or Reciprocity — see **page 4.3a**)
Accuracy: Exact
Reference: **Tada 1985**



$$z = x + iy$$

$$\begin{Bmatrix} Z_I(z) \\ Z_{II}(z) \\ Z_{III}(z) \end{Bmatrix} = \frac{1}{\pi} \begin{Bmatrix} p \\ q \\ t \end{Bmatrix} \left\{ \frac{\sqrt{b^2 - a^2} + f\left(\frac{b}{a}\right) z}{\sqrt{a^2 - z^2}} - \cos^{-1} \frac{bz - a^2}{a(b - z)} \right\}$$

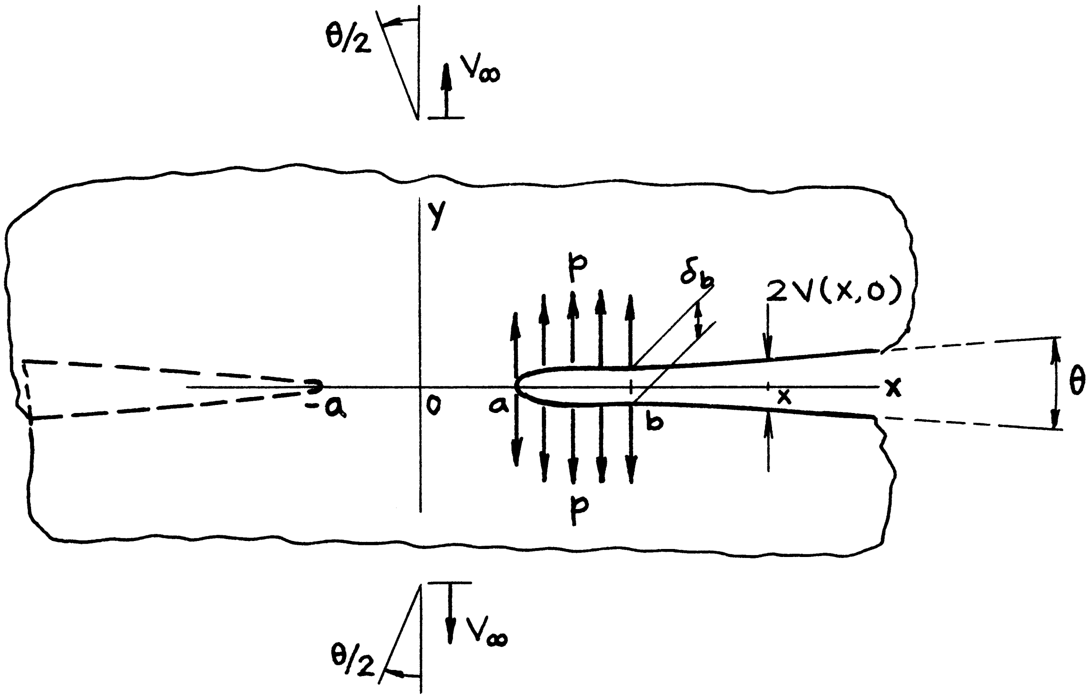
$$\begin{Bmatrix} \overline{Z}_I(z) \\ \overline{Z}_{II}(z) \\ \overline{Z}_{III}(z) \end{Bmatrix} = \frac{1}{\pi} \begin{Bmatrix} p \\ q \\ t \end{Bmatrix} \left\{ -f\left(\frac{b}{a}\right) \sqrt{a^2 - z^2} + (b - z) \sin^{-1} \frac{bz - a^2}{a(b - z)} \right\}$$

$$\begin{Bmatrix} K_I \\ K_{II} \\ K_{III} \end{Bmatrix}_{\pm a} = \frac{1}{\sqrt{\pi a}} \begin{Bmatrix} p \\ q \\ t \end{Bmatrix} \left\{ \sqrt{b^2 - a^2} \pm a f\left(\frac{b}{a}\right) \right\}$$

$$\operatorname{Im} \begin{Bmatrix} \overline{Z}_I(x) \\ \overline{Z}_{II}(x) \\ \overline{Z}_{III}(x) \end{Bmatrix}_{|x| \geq a} = \frac{1}{\pi} \begin{Bmatrix} p \\ q \\ t \end{Bmatrix} \left\{ f\left(\frac{b}{a}\right) \cdot x \sqrt{1 - \left(\frac{a}{x}\right)^2} + (b - x) \cosh^{-1} \left| \frac{bx - a^2}{a(b - x)} \right| \right\}$$

where
$$f\left(\frac{b}{a}\right) = \begin{Bmatrix} \frac{b}{a} \sqrt{\left(\frac{b}{a}\right)^2 - 1} \\ \cosh^{-1} \frac{b}{a} \\ \frac{b}{a} \sqrt{\left(\frac{b}{a}\right)^2 - 1} \end{Bmatrix}$$

Method: Westergaard Stress Function (Integration of **page 4.5**)
Accuracy: Exact
Reference: **Tada 1985**



$$K_{I\pm a} = \frac{p}{\sqrt{\pi a}}\sqrt{b^2 - a^2}\left(1 \pm \frac{b}{a}\right)$$

Crack Opening Profile:

$$\underset{|x|\geq a}{2v(x,0)} = \frac{4p}{\pi E'}\left\{(b-x)\cosh^{-1}\left|\frac{bx-a^2}{a(b-x)}\right| + b\frac{x}{a}\sqrt{\left(\frac{b}{a}\right)^2 - 1}\sqrt{1-\left(\frac{a}{x}\right)^2}\right\}$$

Opening at $x = b$:

$$\delta_b = 2\,v(b,0) = \frac{4pb}{\pi E'}\left\{\ln\frac{b}{a} + \left(\frac{b}{a}\right)^2 - 1\right\}$$

Relative Vertical Displacement along y-Axis at Infinity:

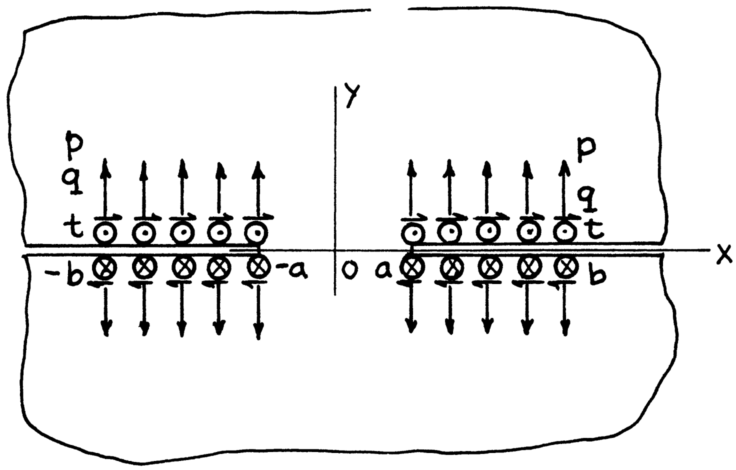
$$2\,v_\infty = \frac{4pb}{\pi E'}\left\{\cosh^{-1}\frac{b}{a} - \sqrt{1-\left(\frac{a}{b}\right)^2}\right\}$$

Relative Rotation at Infinity:

$$\theta = \frac{4p}{\pi E'}\left\{\frac{b}{a}\sqrt{\left(\frac{b}{a}\right)^2 - 1} - \cosh^{-1}\frac{b}{a}\right\}\left(= \left\{\frac{2\,v(x,0)}{x}\right\}_{x\to\infty}\right)$$

Method: Westergaard Stress Function (Integration of **pages 4.5 and 4.5a,b**)
Accuracy: Exact
Reference: **Tada 1985**

NOTE: 1. Always $K_{I-a} < 0$.
2. No surface interference $(x \leq -a)$ was considered.



$$z = x + iy$$

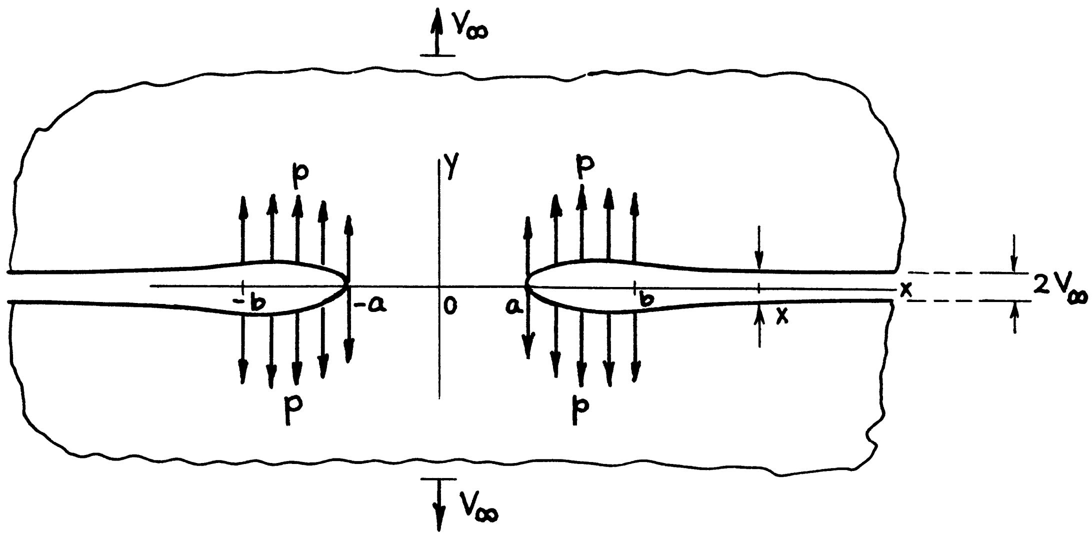
$$\begin{Bmatrix} Z_I(z) \\ Z_{II}(z) \\ Z_{III}(z) \end{Bmatrix} = \frac{2}{\pi} \begin{Bmatrix} p \\ q \\ t \end{Bmatrix} \left\{ \sqrt{\frac{b^2 - a^2}{a^2 - z^2}} - \tan^{-1} \sqrt{\frac{b^2 - a^2}{a^2 - z^2}} \right\}$$

$$\begin{Bmatrix} \overline{Z}_I(z) \\ \overline{Z}_{II}(z) \\ \overline{Z}_{III}(z) \end{Bmatrix} = \frac{2}{\pi} \begin{Bmatrix} p \\ q \\ t \end{Bmatrix} \left\{ b \tan^{-1} \sqrt{\frac{1 - (a/b)^2}{(a/z)^2 - 1}} - z \tan^{-1} \sqrt{\frac{(b/a)^2 - 1}{1 - (z/a)^2}} \right\}$$

$$\begin{Bmatrix} K_I \\ K_{II} \\ K_{III} \end{Bmatrix} = \frac{2}{\sqrt{\pi a}} \begin{Bmatrix} p \\ q \\ t \end{Bmatrix} \sqrt{b^2 - a^2}$$

$$\mathrm{Im} \underset{|x|>a}{\begin{Bmatrix} \overline{Z}_I(x) \\ \overline{Z}_{II}(x) \\ \overline{Z}_{III}(x) \end{Bmatrix}} = \frac{2}{\pi} \begin{Bmatrix} p \\ q \\ t \end{Bmatrix} \left\{ b \begin{pmatrix} \tanh^{-1} \\ \coth^{-1} \end{pmatrix} \cdot \sqrt{\frac{1 - (a/b)^2}{1 - (a/x)^2}} - |x| \begin{pmatrix} \tanh^{-1} \\ \coth^{-1} \end{pmatrix} \sqrt{\frac{(b/a)^2 - 1}{(x/a)^2 - 1}} \right\} \quad \begin{pmatrix} |x| > b \\ a \le |x| < b \end{pmatrix}$$

Method: Integration of **page 4.6** or Superposition of **page 4.7**
Accuracy: Exact
Reference: **Tada 1973**



$$K_{I\pm a} = \frac{2p}{\sqrt{\pi a}}\sqrt{b^2 - a^2}$$

Crack Opening Profile:

$$\underset{|x|\geq a}{2v(x,0)} = \frac{8p}{\pi E'}\left\{ b\begin{pmatrix}\tanh^{-1}\\ \coth^{-1}\end{pmatrix}\sqrt{\frac{1-(a/x)^2}{1-(a/b)^2}} - |x|\begin{pmatrix}\tanh^{-1}\\ \coth^{-1}\end{pmatrix}\sqrt{\frac{(x/a)^2-1}{(b/a)^2-1}}\right\} \quad \begin{pmatrix} a \leq |x| < b \\ x > b\end{pmatrix}$$

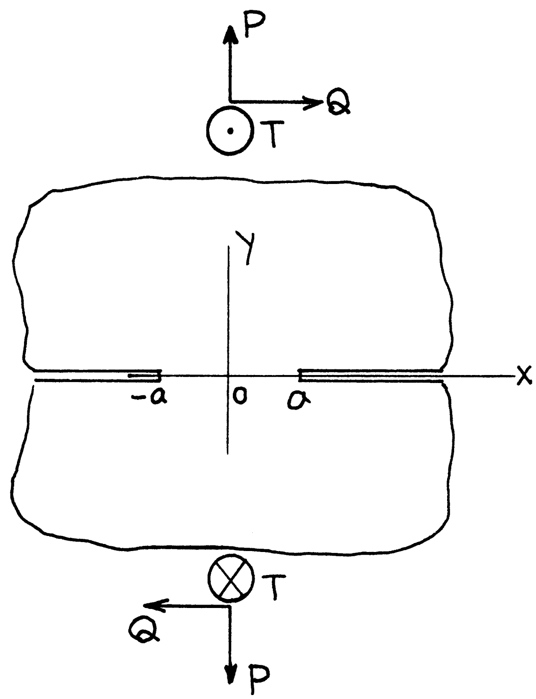
Opening at $|x| = b$:

$$2v(\pm b, 0) = \frac{8pb}{\pi E'}\ell\mathrm{n}\frac{b}{a}$$

Relative Vertical Displacement at Infinity:

$$2v_\infty = \frac{8pb}{\pi E'}\left\{\cosh^{-1}\frac{b}{a} - \sqrt{1-\left(\frac{a}{b}\right)^2}\right\}(= 2v(x\to\infty, 0))$$

Method: Westergaard Stress Function (Integration of **pages 4.6 and 4.6a**)
Accuracy: Exact
Reference: **Tada 1985**



$$z = x + iy$$

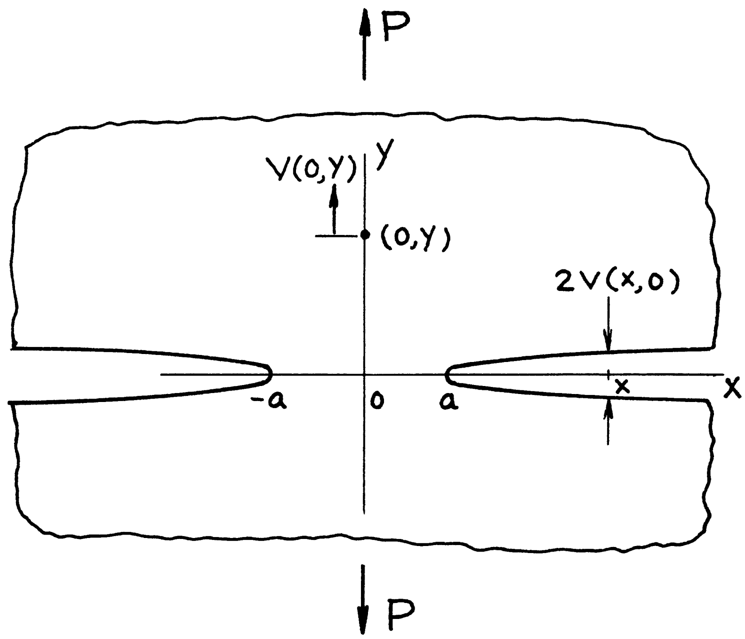
$$\left\{\begin{array}{c} Z_I(z) \\ Z_{II}(z) \\ Z_{III}(z) \end{array}\right\} = \frac{1}{\pi}\left\{\begin{array}{c} P \\ Q \\ T \end{array}\right\}\frac{1}{\sqrt{a^2 - z^2}}$$

$$\left\{\begin{array}{c} \overline{Z}_I(z) \\ \overline{Z}_{II}(z) \\ \overline{Z}_{III}(z) \end{array}\right\} = \frac{1}{\pi}\left\{\begin{array}{c} P \\ Q \\ T \end{array}\right\}\sin^{-1}\frac{z}{a}$$

$$\left\{\begin{array}{c} K_I \\ K_{II} \\ K_{III} \end{array}\right\} = \frac{1}{\sqrt{\pi a}}\left\{\begin{array}{c} P \\ Q \\ T \end{array}\right\}$$

$$\mathrm{Im}\left\{\begin{array}{c} \overline{Z}_I(x) \\ \overline{Z}_{II}(x) \\ \overline{Z}_{III}(x) \end{array}\right\}_{|x|\geq a} = \frac{1}{\pi}\left\{\begin{array}{c} P \\ Q \\ T \end{array}\right\}\cosh^{-1}\frac{x}{a}$$

Methods: Special Case of **page 4.3** or **page 7.1** or by Stress Concentration Factor
Accuracy: Exact
References: **Neuber 1937; Winne 1958; Paris 1960, 1965; Sih 1964**



$$K_I = \frac{P}{\sqrt{\pi a}}$$

Crack Opening Profile:

$$\underset{|x|\geq a}{2\,v(x,0)} = \frac{4P}{\pi E'} \cosh^{-1} \frac{x}{a}$$

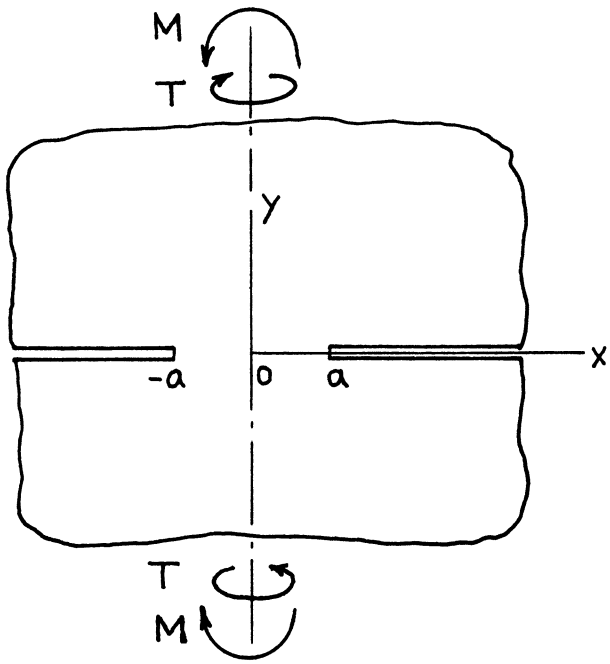
Vertical Displacement at $(0, y)$:

$$v(0,y) = \frac{2P}{\pi E'} \left(\sinh^{-1} \frac{y}{a} - \alpha \frac{y}{\sqrt{a^2 + y^2}} \right)$$

where

$$\alpha = \begin{cases} \frac{1}{2}(1+\nu) & \text{plane stress} \\ \frac{1}{2}\left(\frac{1}{1-\nu}\right) & \text{plane strain} \end{cases}$$

Methods: v Reciprocity (see **pages 4.3a and 4.6a**) or Paris' Equation (see **Appendix B**)
Accuracy: Exact
Reference: **Tada 1985**



$$z = x + iy$$

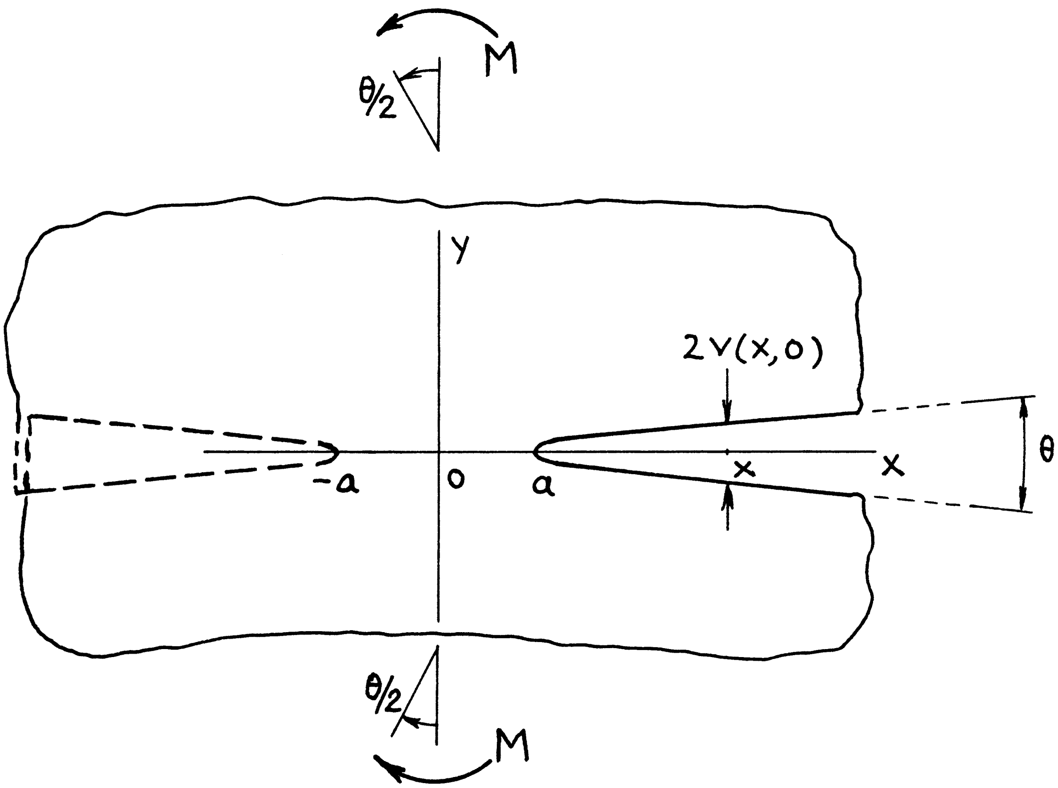
$$\begin{Bmatrix} Z_I(z) \\ Z_{III}(z) \end{Bmatrix} = \frac{2}{\pi a^2} \begin{Bmatrix} M \\ T \end{Bmatrix} \frac{z}{\sqrt{a^2 - z^2}}$$

$$\begin{Bmatrix} \overline{Z}_I(z) \\ \overline{Z}_{III}(z) \end{Bmatrix} = \frac{2}{\pi a^2} \begin{Bmatrix} M \\ T \end{Bmatrix} \left[-\sqrt{a^2 - z^2} \right]$$

$$\begin{Bmatrix} K_I \\ K_{III} \end{Bmatrix}_{\pm a} = \pm \frac{2}{a\sqrt{\pi a}} \begin{Bmatrix} M \\ T \end{Bmatrix}$$

$$\operatorname{Im} \begin{Bmatrix} \overline{Z}_I(x) \\ \overline{Z}_{III}(x) \end{Bmatrix}_{|x| \geq a} = \frac{2}{\pi a^2} \begin{Bmatrix} M \\ T \end{Bmatrix} \cdot x\sqrt{1 - (a/x)^2}$$

Methods: Westergaard Stress Function, etc. or by Stress Concentration Factor
Accuracy: Exact
References: **Neuber 1937; Paris 1960, 1965; Benthem 1972**



$$K_{I\pm a} = \pm\frac{2}{\sqrt{\pi}} \cdot \frac{M}{a^{3/2}} = \pm\frac{\sqrt{\pi}}{4}E'\theta\sqrt{a}$$

Crack Opening Profile:

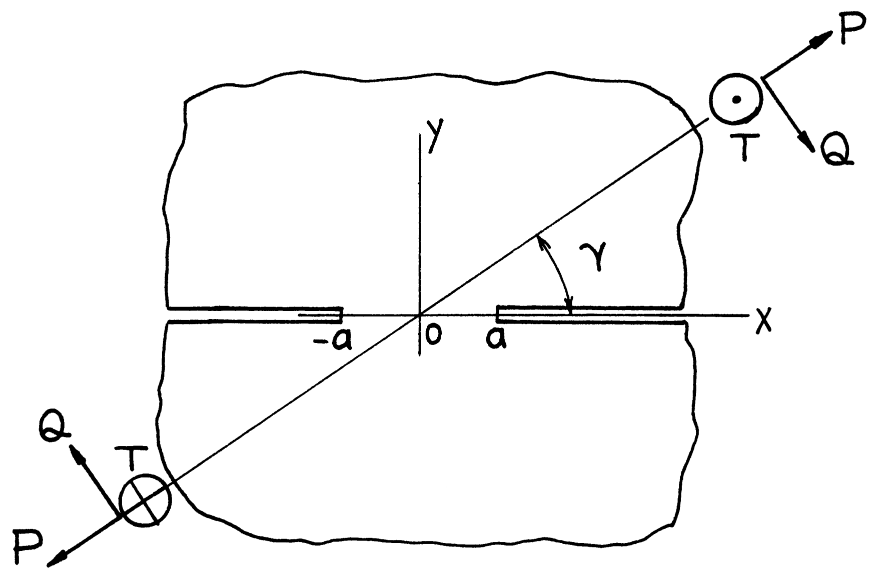
$$\underset{|x|\geq a}{2v(x,0)} = \frac{8M}{\pi E'}\frac{x}{a^2}\sqrt{1-\left(\frac{a}{x}\right)^2}$$

Relative Rotation at Infinity:

$$\theta = \frac{8M}{\pi E' a^2} = \frac{4|K_I|}{\sqrt{\pi}E'\sqrt{a}}\left(=\left\{\frac{2v(x,0)}{x}\right\}_{x\to\infty}\right)$$

Method: v, θ Paris' Equation (see **Appendix B**)
Accuracy: Exact
Reference: **Tada 1985**

NOTE: No surface interference ($x \leq -a$) was considered.



$$z = x + iy$$

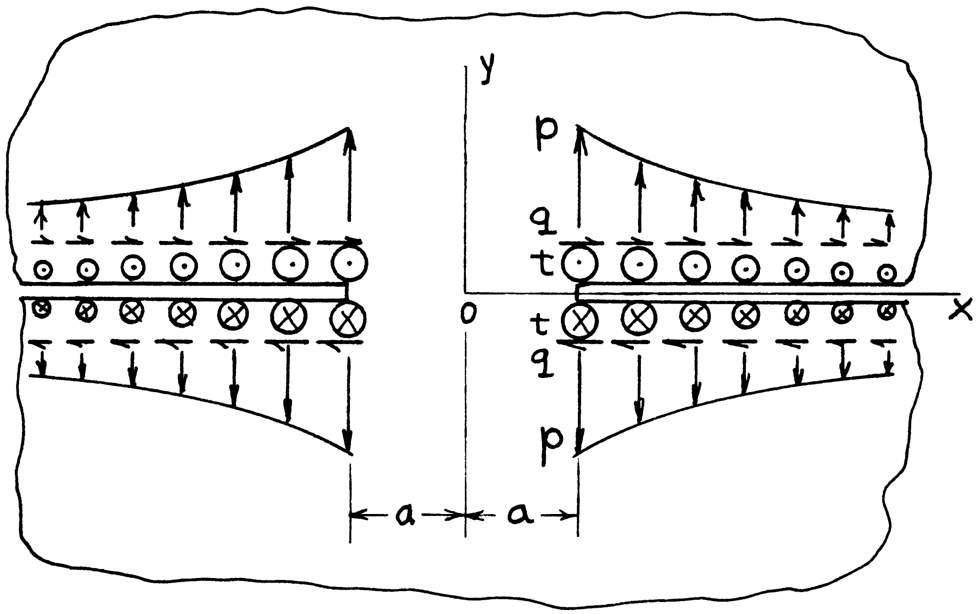
$$\left\{\begin{array}{c} Z_I(z) \\ Z_{II}(z) \\ Z_{III}(z) \end{array}\right\} = \frac{1}{\pi}\left\{\begin{array}{c} P\sin\gamma - Q\cos\gamma \\ P\cos\gamma + Q\sin\gamma \\ T \end{array}\right\} \frac{1}{\sqrt{a^2 - z^2}}$$

$$\left\{\begin{array}{c} \overline{Z}_I(z) \\ \overline{Z}_{II}(z) \\ \overline{Z}_{III}(z) \end{array}\right\} = \frac{1}{\pi}\left\{\begin{array}{c} P\sin\gamma - Q\cos\gamma \\ P\cos\gamma + Q\sin\gamma \\ T \end{array}\right\} \sin^{-1}\frac{z}{a}$$

$$\left\{\begin{array}{c} K_I \\ K_{II} \\ K_{III} \end{array}\right\} = \frac{1}{\sqrt{\pi a}}\left\{\begin{array}{c} P\sin\gamma - Q\cos\gamma \\ P\cos\gamma + Q\sin\gamma \\ T \end{array}\right\}$$

$$\operatorname{Im}\underset{|x|\geq a}{\left\{\begin{array}{c} \overline{Z}_I(x) \\ \overline{Z}_{II}(x) \\ \overline{Z}_{III}(x) \end{array}\right\}} = \frac{1}{\pi}\left\{\begin{array}{c} P\sin\gamma - Q\cos\gamma \\ P\cos\gamma + Q\sin\gamma \\ T \end{array}\right\} \cosh^{-1}\frac{x}{a}$$

Method: Superposition of **page 4.9**
Accuracy: Exact
References: **Neuber 1937; Winne 1958; Paris 1960, 1965; Sih 1964**

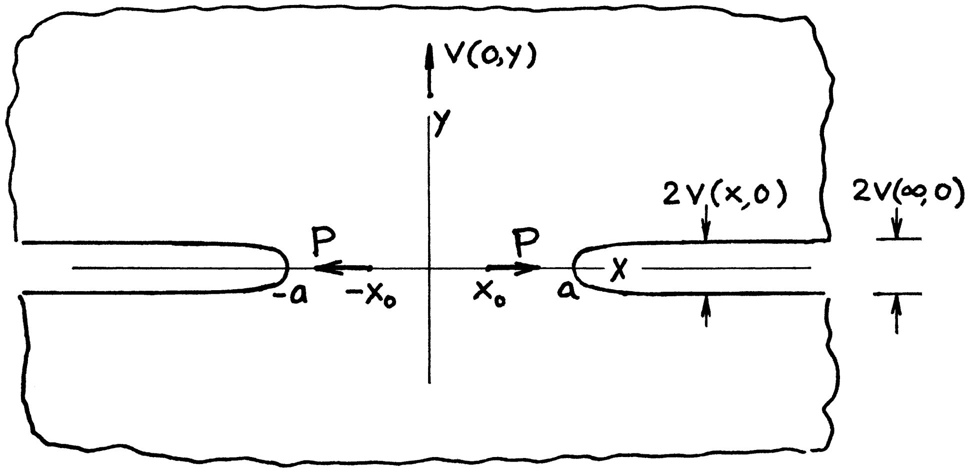


$$\left\{\begin{array}{c}\sigma_y(x,0)\\ \tau_{xy}(x,0)\\ \tau_{yz}(x,0)\end{array}\right\}_{|x|\geq a} = \left\{\begin{array}{c}p\\ q\\ t\end{array}\right\}\left|\frac{a}{x}\right|^{\gamma} \quad (\gamma > 1)$$

$$\left\{\begin{array}{c}K_I\\ K_{II}\\ K_{III}\end{array}\right\} = \left\{\begin{array}{c}p\\ q\\ t\end{array}\right\}\sqrt{a}\frac{\Gamma\left(\frac{\gamma-1}{2}\right)}{\Gamma\left(\frac{\gamma}{2}\right)}$$

where $\Gamma(\gamma) =$ Gamma Function (See **Appendix M**)

Method: Integration of **page 4.6**
Accuracy: Exact
Reference: **Tada 1974**



$$K_I = \frac{P(1-\alpha)}{\sqrt{\pi a}} \frac{x_0}{\sqrt{a^2 - x_0^2}}$$

Crack Opening Profile:

$$\underset{|x|\geq a}{2v(x,0)} = \frac{8P(1-\alpha)}{\pi E'} \cos^{-1} \sqrt{\frac{1-(x_0/a)^2}{1-(x_0/x)^2}}$$

$$2v(\infty,0) = \frac{8P(1-\alpha)}{\pi E'} \sin^{-1} \frac{x_0}{a}$$

Displacement at $(0,y)$:

$$v(0,y) = \frac{4P(1-\alpha)}{\pi E'} \left(1 - \alpha y \frac{\partial}{\partial y}\right) \left\{ \sin^{-1} \sqrt{\frac{(y/a)^2+1}{(y/b)^2+1}} - \tan^{-1} \frac{x_0}{y} \right\}$$

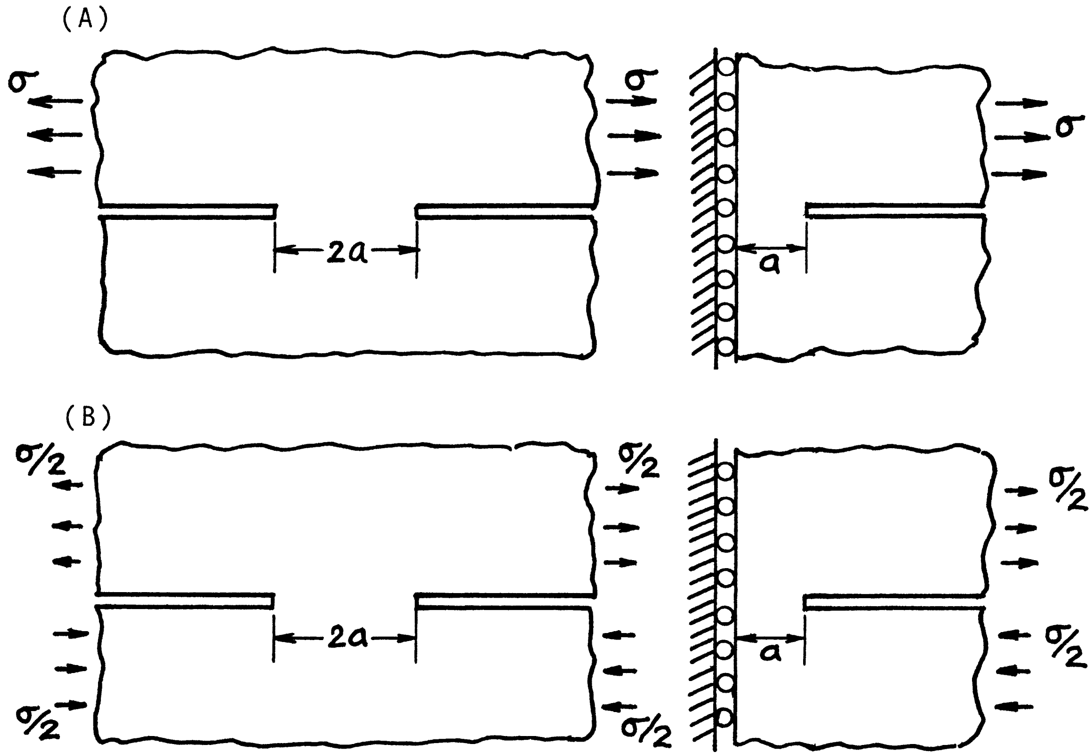
Relative Vertical Displacement at Infinity:

$$2v_\infty = v(0,\infty) - v(0,-\infty) = 2v(\infty,0)$$

where

$$\alpha = \begin{cases} \frac{1}{2}(1+\nu) & \text{plane stress} \\ \frac{1}{2}\left(\frac{1}{1-\nu}\right) & \text{plane strain} \end{cases}$$

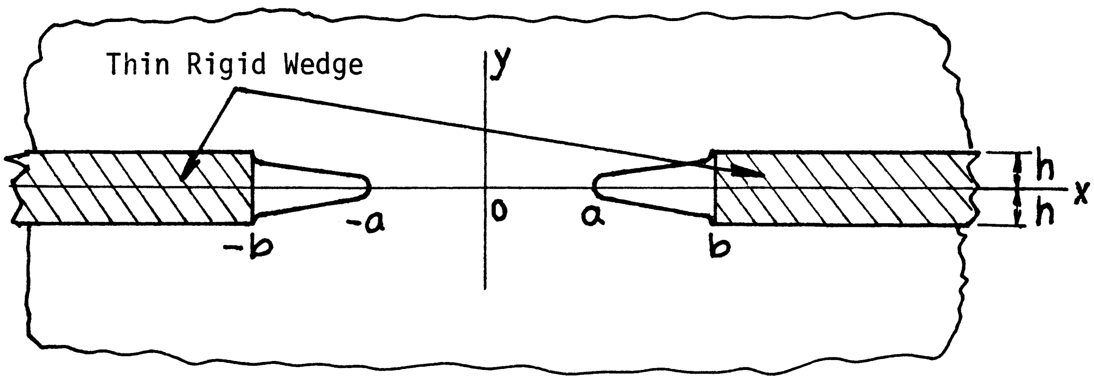
Method: Integration of **page 4.6**; Paris' Equation (see **Appendix B**)
Accuracy: Exact
Reference: **Tada 2000**



$$K_I = 0$$

$$K_{II} = \frac{1}{4}\sigma\sqrt{\pi a}$$

Method: (A) Special (Limiting) Case of **page 21.1** or **21.3**; (B) Superposition of (A)
Accuracy: Exact
Reference: **Tada 2000**



$$K_{I}\underset{x=\pm a}{} = \frac{E'h}{2kK(k)}\sqrt{\frac{\pi}{a}}$$

$$K_{I}\underset{x=\pm b}{} = -\frac{E'h}{2kK(k)}\sqrt{\frac{\pi}{b}}$$

$$\underset{a\le|x|\le b}{2v(x,0)} = 2h\left\{1-\frac{F(\varphi,k)}{K(k)}\right\}$$

where

$$E' = \begin{cases} E & \text{plane stress} \\ E/(1-\nu^2) & \text{plane strain} \end{cases}$$

$$k = \sqrt{1-(a/b)^2}$$

$$\varphi = \sin^{-1}\sqrt{\frac{b^2-x^2}{b^2-a^2}}$$

$$F(\varphi, k) = \int_0^{\varphi} \frac{d\varphi}{\sqrt{1 - k^2 \sin^2 \varphi}}$$

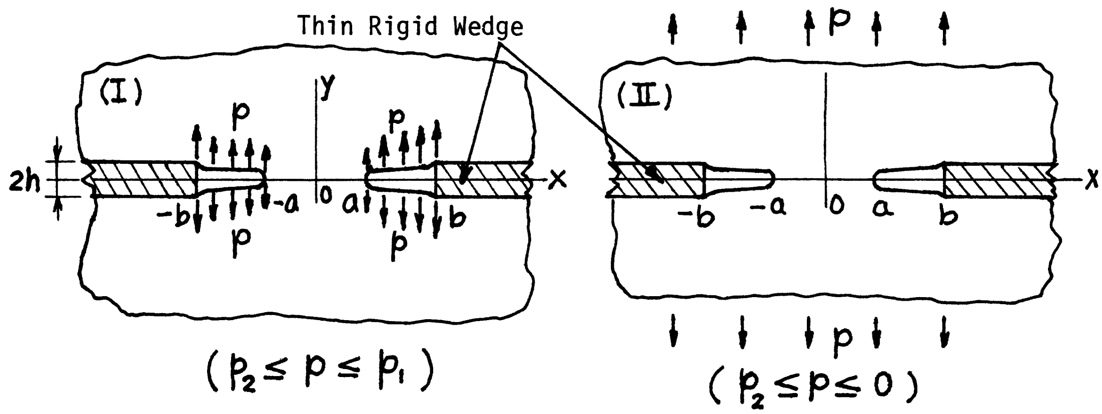
$$K(k) = F(\pi/2, k)$$

Method: Westergaard Stress Function (or Negative of **page 5.21**)
Accuracy: Exact
Reference: **Tada 1974**

NOTE: For $K(k)$, see **Appendix L**.

The Westergaard Function is

$$Z_I(z) = -\frac{E'hb}{2K(k)} \cdot \frac{1}{\sqrt{z^2 - a^2}} \frac{1}{\sqrt{z^2 - b^2}}$$



$$p_1 = \frac{E'h}{2b\{K(k) - E(k)\}}$$

$$p_2 = -\frac{E'hb}{2\left\{b^2E(k) - a^2K(k)\right\}}$$

$$\underset{x=\pm a}{K_I} = \left\{\frac{E'h}{2} + p\frac{b^2E(k) - a^2K(k)}{b}\right\}\frac{1}{kK(k)}\sqrt{\frac{\pi}{a}}$$

$$\underset{x=\pm b}{K_I} = \left\{-\frac{E'h}{2} + pb(K(k) - E(k))\right\}\frac{1}{kK(k)}\sqrt{\frac{\pi}{b}}$$

$$\underset{a\leq|x|\leq b}{2v(x,0)} = 2h\left\{1 - \frac{F(\varphi, k)}{K(k)}\right\} + \frac{4pb}{E'}\{K(k)E(\varphi, k) - E(k)F(\varphi, k)\}$$

where

$$E' = \begin{cases} E & \text{plane stress} \\ E/(1-\nu^2) & \text{plane strain} \end{cases}$$

$$k = \sqrt{1 - (a/b)^2}, \quad \varphi = \sin^{-1}\sqrt{\frac{b^2 - x^2}{b^2 - a^2}}$$

$$F(\varphi, k) = \int_0^{\varphi} \frac{d\varphi}{\sqrt{1 - k^2 \sin^2 \varphi}}, \quad K(k) = F(\pi/2, k)$$

$$E(\varphi, k) = \int_0^{\varphi} \sqrt{1 - k^2 \sin^2 \varphi}\, d\varphi, \quad E(k) = E(\pi/2, k)$$

Method: Superposition of **page 4.14** and **page 6.1**
Accuracy: Exact
Reference: **Tada 1974**

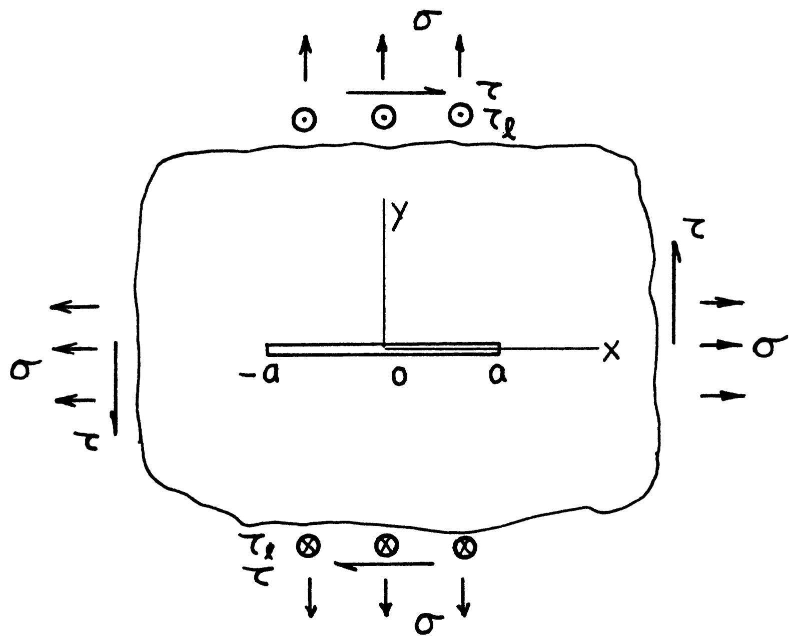
NOTE:

1. In (I), when $p > p_1$, separation of contact surfaces occurs near $x = \pm b$, and when $p < p_2$, crack closure occurs near $x = \pm a$.

2. In (II), when p > 0 (remote tension), separation of contact surfaces occurs for large $|x|$, and when $p < p_2$, crack closure occurs near $x = \pm a$.

3. For $K(k)$ and $E(k)$, see **Appendix L.**

The Westergaard Function is $Z_I(z) = Z_I(z) + Z_I(z)$
p.4.16 **p.4.15** **p.6.1** (replace σ by p)



$$z = x + iy$$

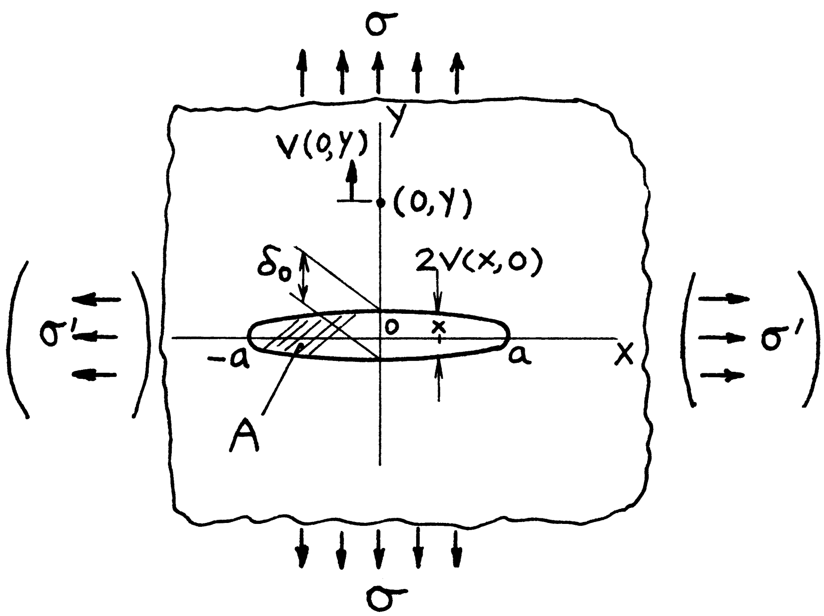
$$\begin{Bmatrix} Z_I(z) \\ Z_{II}(z) \\ Z_{III}(z) \end{Bmatrix} = \begin{Bmatrix} \sigma \\ \tau \\ \tau_\ell \end{Bmatrix} \frac{z}{\sqrt{z^2 - a^2}}$$

$$\begin{Bmatrix} \overline{Z}_I(z) \\ \overline{Z}_{II}(z) \\ \overline{Z}_{III}(z) \end{Bmatrix} = \begin{Bmatrix} \sigma \\ \tau \\ \tau_\ell \end{Bmatrix} \sqrt{z^2 - a^2}$$

$$\begin{Bmatrix} K_I \\ K_{II} \\ K_{III} \end{Bmatrix} = \begin{Bmatrix} \sigma \\ \tau \\ \tau_\ell \end{Bmatrix} \sqrt{\pi a}$$

$$\mathrm{Im} \begin{Bmatrix} \overline{Z}_I(x) \\ \overline{Z}_{II}(x) \\ \overline{Z}_{III}(x) \end{Bmatrix}_{|x| \le a} = \begin{Bmatrix} \sigma \\ \tau \\ \tau_\ell \end{Bmatrix} \sqrt{a^2 - x^2}$$

Methods: Westergaard Stress Function, etc. or by Stress Concentration Factor
Accuracy: Exact
References: **Griffith 1920; Westergaard 1939; Irwin 1957, 1958a; Paris 1965, etc.**



$$K_I = \sigma\sqrt{\pi a}$$

Crack Opening Area:

$$A = \frac{2\sigma\pi a^2}{E'}$$

Crack Opening Profile:

$$\underset{|x|\le a}{2v(x,0)} = \frac{4\sigma}{E'}\sqrt{a^2 - x^2}$$

Opening at Center:

$$\delta_0 = 2v(0,0) = \frac{4\sigma a}{E'}$$

Additional Vertical Displacement at $(0, y)$ due to Crack:

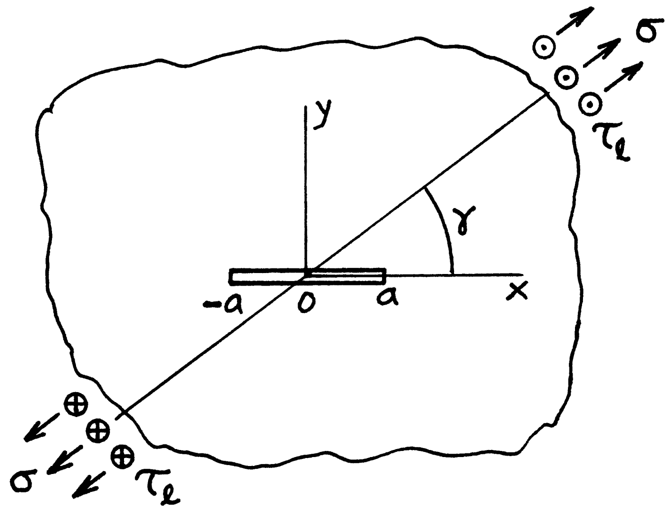
$$v(0,y) = \frac{2\sigma}{E'}\left(\sqrt{a^2+y^2} - y\right)\left(1 + \alpha\frac{y}{\sqrt{a^2+y^2}}\right)$$

where

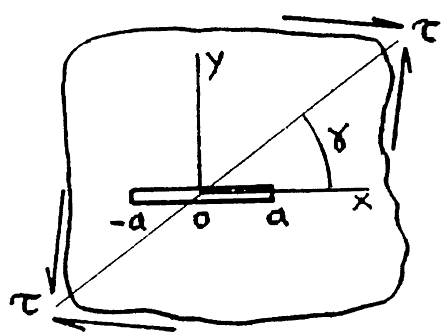
$$\alpha = \begin{cases} \frac{1}{2}(1+\nu) & \text{plane stress} \\ \frac{1}{2}\left(\frac{1}{1-\nu}\right) & \text{plane strain} \end{cases}$$

Method: Paris' Equation (see **Appendix B**)
Accuracy: Exact
Reference: **Tada 1985**

NOTE: $v(0, y)$ is the displacement at $(0, y)$ when uniform pressure σ is applied on crack surfaces.



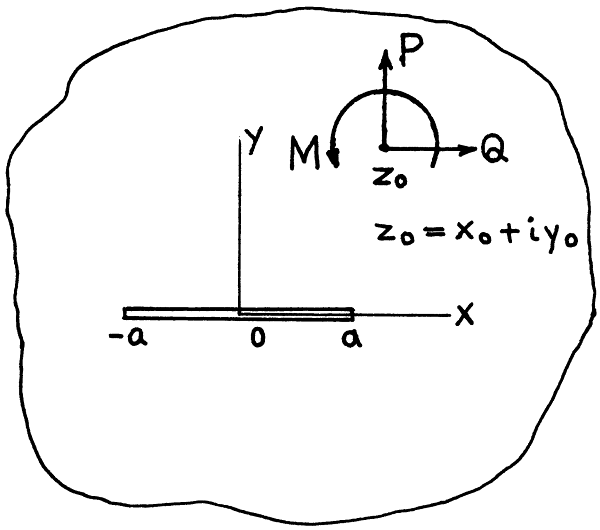
$$\begin{Bmatrix} K_I \\ K_{II} \\ K_{III} \end{Bmatrix} = \begin{Bmatrix} \sigma \sin\gamma \\ \sigma\cos\gamma \\ \tau_\ell \end{Bmatrix} \sin\gamma \cdot \sqrt{\pi a}$$



$$\begin{Bmatrix} K_I \\ K_{II} \end{Bmatrix} = \tau \begin{Bmatrix} -\cos 2\gamma \\ \sin 2\gamma \end{Bmatrix} \cdot \sqrt{\pi a}$$

$$K_{II} = 0$$

Method: Superposition
Accuracy: Exact
References: **Paris 1965; Sih 1965a**



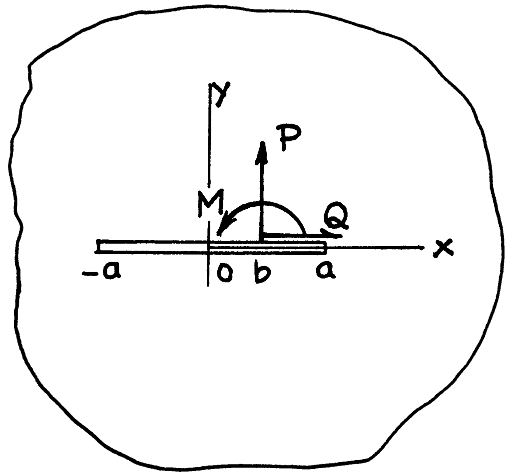
$$K_{+a} = (K_I - iK_{II})_{+a}$$

$$= \frac{1}{2\sqrt{\pi a}} \frac{1}{\kappa+1} \left\{ (Q+iP) \left[\left(\frac{a+z_0}{\sqrt{z_0^2-a^2}} - 1 \right) - \kappa \left(\frac{a+\bar{z}_0}{\sqrt{\bar{z}_0^2-a^2}} - 1 \right) \right] + \frac{a[(Q-iP)(\bar{z}_0 - z_0)+i(1+\kappa)M]}{(\bar{z}_0-a)\sqrt{\bar{z}_0^2-a^2}} \right\}$$

where

$$\kappa = \begin{cases} \frac{3-\nu}{1+\nu} & \text{plane stress} \\ 3-4\nu & \text{plane strain} \end{cases}$$

Method: Muskhelishvili's Method
Accuracy: Exact
References: **Erdogan 1962; Sih 1962a**



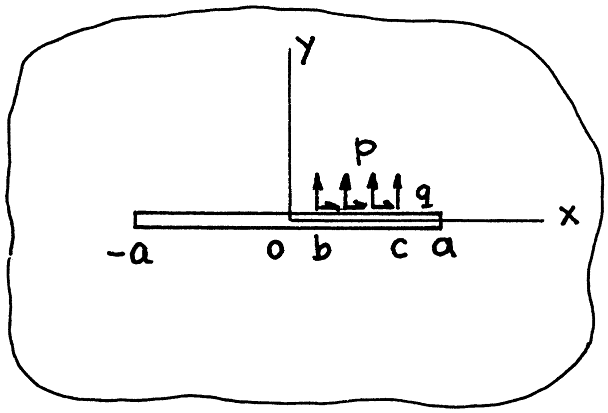
$$K_{I}_{+a} = \frac{1}{2\sqrt{\pi a}}\left\{P\sqrt{\frac{a+b}{a-b}} + \left(\frac{\kappa-1}{\kappa+1}\right)Q + \frac{Ma}{(a-b)\sqrt{a^2-b^2}}\right\}$$

$$K_{II}_{+a} = \frac{1}{2\sqrt{\pi a}}\left\{Q\sqrt{\frac{a+b}{a-b}} - \left(\frac{\kappa-1}{\kappa+1}\right)P\right\}$$

where

$$\kappa = \begin{cases} \frac{3-\nu}{1+\nu} & \text{plane stress} \\ 3-4\nu & \text{plane strain} \end{cases}$$

Method: Muskhelishvili's Method
Accuracy: Exact
References: **Erdogan 1962; Sih 1962a**



$$K_{I\pm a} = \frac{1}{2}\,\frac{1}{\sqrt{\pi a}}\left[pa\left(\sin^{-1}\frac{c}{a} - \sin^{-1}\frac{b}{a} \mp \sqrt{1-(c/a)^2} \pm \sqrt{1-(b/a)^2}\right) + \left(\frac{\kappa-1}{\kappa+1}\right)q(c-b)\right]$$

$$K_{II\pm a} = \frac{1}{2}\frac{1}{\sqrt{\pi a}}\left[qa\left(\sin^{-1}\frac{c}{a} - \sin^{-1}\frac{b}{a} \mp \sqrt{1-(c/a)^2} \pm \sqrt{1-(b/a)^2}\right) - \left(\frac{\kappa-1}{\kappa+1}\right)p(c-b)\right]$$

where

$$\kappa = \begin{cases} \frac{3-\nu}{1+\nu} & \text{plane stress} \\ 3-4\nu & \text{plane strain} \end{cases}$$

Method: Muskhelishvili’s Method (Integration of **page 5.4**)
Accuracy: Exact
References: **Erdogan 1962; Sih 1962a**